Design, Verification, and Comparative Performance Evaluation of a Scalable 8×8 and 16×16 MIMO-OFDM PHY Simulator with Threshold-Based Adaptive Modulation and Fixed-Rate Channel Coding

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Abstract

Hands-on algorithm competence across the full physical-layer receiver, synchronization, channel estimation, MIMO processing, equalization, and link adaptation, is best demonstrated by a complete, executed, and verified simulation chain rather than by isolated block studies. This paper presents a MIMO-OFDM link-level simulator implemented in MATLAB and exercised at two square array sizes, 8×8 and 16×16, under otherwise identical conditions, so that the antenna count is the only changed variable and every difference between the two result sets is attributable to array size alone. Across the two coordinated studies, the package implements even-bin Schmidl-Cox timing acquisition with coarse and fine carrier-frequency-offset recovery, least-squares and statistics-aware Wiener MMSE channel estimation over orthogonal per-antenna pilot combs, zero-forcing and soft-output unbiased MMSE detection with per-stream effective noise variance, and threshold-based adaptive modulation over a fixed rate-1/2 convolutional code, adaptive modulation with a fixed-rate code rather than complete adaptive modulation and coding, decoded by both hard-decision and soft-decision Viterbi algorithms. Hard bits and soft log-likelihood ratios are generated from the same gain-corrected unbiased MMSE symbol stream, ensuring that the measured decoder separation is attributable to the use of hard or soft reliability information. Every algorithm is presented as a formal derivation, as a numbered procedure with a workflow diagram, and as a traceability mapping to the implementing MATLAB function. Verification followed an execution-first discipline that exposed four genuine defects, a corrupted genie synchronization reference, a missing FFT processing-gain factor of 24 dB in the noise calibration, and a preamble subcarrier-parity error that displaced the coarse carrier estimate by exactly one subcarrier spacing while leaving the timing metric deceptively healthy, and an inconsistent MMSE stream in the original hard-and-soft decoder comparison, where the hard branch used the amplitude-biased output while the soft branch used the gain-corrected unbiased output; each defect was isolated by an independent numerical experiment, corrected, and confirmed by rerun. The verified campaigns measure a recovered synchronization chain overlapping the genie reference with a carrier-offset RMSE of 10^{-3} subcarrier spacings at 30 dB, an ergodic-capacity doubling from 69.3 to 138.0 bit/s/Hz at the array doubling, a least-squares estimation floor emerging at the halved pilot density while the Wiener estimator retains its slope, a zero-forcing detector that never reaches positive output SINR at 16×16 while unbiased MMSE concedes about 1.1 dB of output SINR to the array doubling, and a soft-over-hard Viterbi separation, measured from the common unbiased stream at a coded BER of 10^{-1} and reported with the executed results, that supports the central finding that the value of statistical processing grows with array size. All reported numbers derive from stored per-run result files, and refusals to extrapolate beyond the measured sweep are reported as findings rather than hidden.

Keywords: MIMO-OFDM, link-level simulation, Schmidl-Cox synchronization, Wiener channel estimation, MMSE equalization, soft-decision Viterbi, adaptive modulation, verification methodology.

1. Introduction

The performance of a wireless modem emerges from a chain of tightly coupled processing stages, and none of its algorithms can be evaluated in isolation: a synchronization error corrupts every estimate downstream, an estimation error shifts every detector decision, and a miscalibrated reliability value silently degrades every soft decoder. Link-level simulation is therefore the standard evaluation method of physical-layer engineering, exercising the complete chain under controlled and repeatable conditions. Two practical weaknesses recur in such studies. The first concerns correctness: a simulator is itself a piece of engineering, and a subtle defect in noise calibration, transform scaling, or index arithmetic shifts every curve it produces, yet results are frequently presented without documenting how the implementation was verified. The second concerns comparability: when configurations are compared, differences in secondary parameters contaminate the attribution of the observed differences to the parameter of interest.

This paper addresses both weaknesses in the context of a complete MIMO-OFDM physical-layer simulator implemented in MATLAB and executed at two square array sizes, 8×8 and 16×16. The two coordinated studies together cover the five algorithm areas of a modern PHY/modem receiver. Synchronization comprises Schmidl-Cox timing acquisition [1] together with coarse and fine carrier-frequency-offset estimation and compensation. Channel estimation comprises a least-squares estimator with spline interpolation and a statistics-aware Wiener MMSE estimator in the tradition of van de Beek et al. [2] and Edfors et al. [3], operating over orthogonal per-antenna pilot combs [4]. The MIMO area comprises spatial multiplexing over a time-varying multipath channel and singular-value ergodic capacity after Telatar [5]. Equalization comprises per-subcarrier zero-forcing and unbiased MMSE detection [8], [10], extended to a soft-output formulation with per-stream effective noise variance. Link adaptation comprises threshold-based adaptive modulation [9] over a fixed rate-1/2 convolutional code, adaptive modulation with a fixed-rate code rather than complete adaptive modulation and coding, decoded by the Viterbi algorithm [11] in both hard-decision and soft-decision form [12], with hard bits and soft log-likelihood ratios generated from the same gain-corrected unbiased MMSE symbol stream, so that the measured decoder separation is attributable to the use of hard or soft reliability information.

Comparability is enforced by construction: the two configurations share the channel profile, the Doppler model, the SNR grid, the modulation thresholds, the channel code, the Monte Carlo depth, and the random seed, so the antenna count is the only changed variable. Correctness is established by an execution-first verification discipline under which the chain was run, every metric was compared against its theoretically required behavior, and every inconsistency was isolated by an independent numerical experiment before any code was changed; four genuine defects were found, corrected, and confirmed by rerun, and their record forms a dedicated section of this paper. Every reported number derives from stored per-run result files, and the analysis explicitly refuses to extrapolate beyond the measured sweep.

1.1 Contributions and paper organization

The contributions are threefold. First, the paper documents a five-area PHY/modem simulation package comprising two coordinated studies in three aligned forms, a formal derivation of every algorithm, a numbered procedure with a workflow diagram, and a traceability mapping to the implementing MATLAB function, so that the mathematics, the code, and the reported results remain aligned throughout. Second, the paper executes a controlled array-size study in which the 8×8 and 16×16 configurations differ in nothing but the antenna count, and quantifies from stored result files how each receiver algorithm class responds to the doubling: capacity doubles, interpolation-based estimation hits a floor, exact-inversion detection loses its remaining margin, and the soft-decision gain, measured from the common unbiased stream, grows with the array size. Third, the paper publishes its verification record, four defects with symptom, independent numerical diagnosis, correction, and rerun confirmation, treating the establishment of correctness as a reportable result rather than an assumed precondition.

The remainder of the paper is organized as follows. Section 2 states the system model and the shared simulation parameters. Section 3 derives the receiver algorithms and presents them as numbered procedures with their workflow diagrams and their implementation traceability. Section 4 describes the simulator architecture and its result-provenance chain. Section 5 documents the verification methodology and the defect record. Section 6 reports the executed results of both configurations, Section 7 presents the controlled size comparison, Section 8 states the limitations and future work, and Section 9 concludes.

2. System Model and Simulation Parameters

Section 2 states the system model in the order in which the implementation executes it and collects the simulation parameters shared by the two configurations. Section 2.1 defines the OFDM processing and the multipath channel, Section 2.2 defines the temporal variation and the power convention, and Section 2.3 states the shared parameter set and its reproducibility rules.

2.1 OFDM processing and the multipath channel

All processing takes place in the discrete-time complex baseband domain. The link operates over an OFDM grid of 256 subcarriers with a cyclic prefix of 20 samples at a 30.72 MHz sample rate, giving an OFDM symbol length of 276 samples. The multipath channel follows a five-tap TDL-A style delay profile in the spirit of the 3GPP tapped-delay-line models [14], with integer tap delays of 0, 1, 2, 4, and 6 samples and tap powers of 0.0, −2.2, −4.0, −6.5, and −9.0 dB normalized to unit total power; each tap gain is an independent complex Gaussian draw, giving Rayleigh fading per path. The maximum tap delay lies well inside the cyclic prefix, so cyclic-prefix removal converts the linear channel convolution into a per-subcarrier multiplication and every receiver algorithm of Section 3 operates subcarrier by subcarrier on that basis.

2.2 Temporal variation and the power convention

Temporal variation follows a simplified common-Doppler phase-evolution model [13] at a terminal velocity of 30 km/h and a carrier frequency of 3.5 GHz, a maximum Doppler frequency of approximately 97 Hz, with the channel matrix rebuilt at every OFDM symbol time so that the data symbols late in the slot see a channel that has drifted from the pilot-time channel used for estimation. One power convention governs every SNR-dependent result. The QAM constellations of all orders are scaled to unit average symbol power, the transmit power is split equally across the antennas so that the detector-side channel carries a $1/\sqrt{N_t}$ scaling, and the noise variance is fixed per SNR point independently of the channel realization. Comparisons between the two array sizes are therefore made at equal total transmit power, and every comparative gain reported in Section 7 is a genuine error-rate improvement rather than a hidden power increase.

2.3 Shared parameters and reproducibility

Table 1 collects the parameters shared by both configurations.

Table 1. Shared simulation parameters, held identical across the 8×8 and 16×16 configurations.

Parameter	Value
FFT size / subcarriers	256
Cyclic prefix	20 samples (symbol length 276 at 30.72 MHz)
Channel profile	5-tap TDL-A style, delays 0/1/2/4/6 samples, powers 0/−2.2/−4/−6.5/−9 dB, Rayleigh taps
Doppler model	Common-Doppler phase evolution, 30 km/h at 3.5 GHz (f_D ≈ 97 Hz), channel rebuilt per OFDM symbol
SNR sweep	0 to 30 dB in 2 dB steps, total transmit SNR
Pilot design	Orthogonal comb, one non-overlapping comb per transmit antenna (32 pilots/antenna at 8×8, 16 at 16×16)
Adaptive modulation	QPSK for SNR < 8 dB, 16-QAM for 8 ≤ SNR < 18 dB, 64-QAM for SNR ≥ 18 dB
Channel code	Rate-1/2 convolutional, constraint length 7, generators [133 171], traceback depth 35
Monte Carlo depth	400 trials/SNR point (synchronization study), 100 slots/SNR point (MIMO study), 13 data symbols/slot
Reproducibility	Fixed seed rng(7); preamble drawn under saved-and-restored generator state; results exported to CSV

The first column of Table 1 lists the configured aspects of the link, and the second column states the executed value together with its size-dependent consequence where one exists. Two rows deserve emphasis. The pilot-design row is the mechanism of the estimation results of Section 7: the same 256-subcarrier grid divides among the transmit antennas, so doubling the array halves the per-antenna pilot density from a spacing of 8 subcarriers to a spacing of 16. The reproducibility row is the provenance foundation of the paper: both main scripts run under one fixed seed, the known preamble sequence is drawn under a saved-and-restored generator state so that it does not perturb the Monte Carlo stream, and every aggregated result is written to a per-configuration CSV file from which the comparison figures and the quoted crossings are generated.

3. Receiver Algorithms: Derivations, Procedures, and Traceability

Section 3 derives the receiver algorithms of the five areas and presents each in three aligned forms: a formal derivation in which every symbol is defined, a workflow diagram of the processing steps, and a boxed numbered procedure followed by a description of its flow. Section 3.1 covers timing acquisition and carrier recovery, Section 3.2 the two channel estimators, Section 3.3 the two detectors and the soft-output formulation, Section 3.4 link adaptation and the dual-decoder coding study, Section 3.5 the performance metrics, and Section 3.6 the end-to-end procedure with the traceability table.

3.1 Schmidl-Cox timing acquisition and carrier-frequency-offset recovery

Timing acquisition uses a Schmidl-Cox preamble [1], one OFDM symbol whose QAM training symbols occupy only the even frequency bins in the zero-based sense; energy restricted to even bins makes the two halves of the time-domain symbol body identical, and that repetition is the property the receiver exploits. Writing $L = N/2 = 128$ for the half-symbol length and $r[n]$ for the received samples, the receiver forms a sliding half-symbol autocorrelation and a sliding energy term over the search range,

$$M(d) = \Sigma_k\, r^*(d+k)\, r(d+k+L)\,, \quad P(d) = \Sigma_k\, |r(d+k+L)|^2 \tag{1}$$

where the sums run over $k = 0, \ldots, L-1$, and normalizes their squared magnitudes into the timing metric

$$\Lambda(d) = |M(d)|^2 \,/\, P(d)^2\,. \tag{2}$$

The metric approaches unity when the window straddles the two identical halves and falls away elsewhere; the timing estimate $\hat{d}$ is the position of its maximum. Both sums are evaluated over the whole range by cumulative sums in schmidl_cox_metric.m, which removes the inner loop and keeps the search cost linear in the range length. A documented property of the original construction is a plateau of the metric across the cyclic-prefix region rather than a sharp peak, whose statistical consequence is quantified with the results of Section 6.

A carrier frequency offset of ε subcarrier spacings advances the phase between the two halves by $\pi\varepsilon$, so the coarse estimate follows directly from the autocorrelation angle at the detected timing,

$$\hat{\varepsilon}\_coarse = angle(\, M(\hat{d})\,) \,/\, \pi\,, \tag{3}$$

with an unambiguous range of one subcarrier spacing in each direction, implemented by estimate_coarse_cfo.m. After coarse compensation by the opposite phase ramp, the residual offset is estimated from the pilot phase drift between two consecutive data OFDM symbols. With identical unit pilots in both symbols the channel contribution cancels in the pilot cross-correlation, and the per-symbol rotation converts to a normalized offset through the symbol length $N\_sym = N + N\_CP$,

$$\hat{\varepsilon}\_fine = angle(\, \Sigma_p\, Y_2(p)\, Y_1{}^*(p)\,) \cdot N \,/\, (\, 2\pi\, N\_sym\,)\,, \tag{4}$$

implemented by estimate_fine_cfo.m, where $Y_1(p)$ and $Y_2(p)$ denote the pilot-subcarrier observations of the two symbols. The receiver compensates with the sum of the two estimates. The evaluation of Section 6 compares three demodulation cases at every trial: a genie reference in which the received signal is compensated with the true offset, an uncompensated case that quantifies the inter-carrier-interference floor, and the recovered case; the true transmitted offset is 0.25 subcarrier spacings throughout. Figure 1 presents the synchronization pipeline as a workflow diagram.

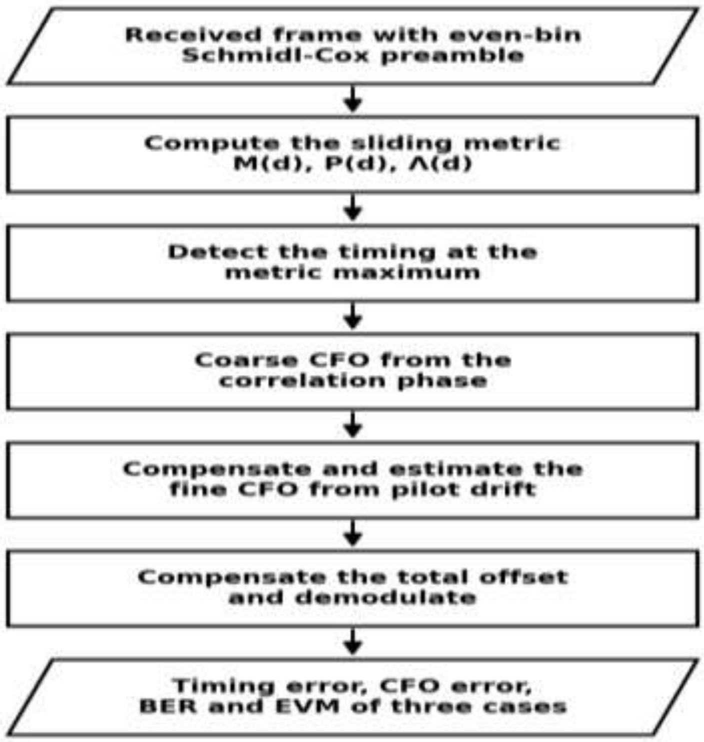

Figure 1. Synchronization workflow: even-bin Schmidl-Cox preamble, cumulative-sum timing metric, coarse carrier recovery from the correlation phase, and fine recovery from pilot drift, equations (1)–(4).

Algorithm 1: Schmidl-Cox timing acquisition and two-stage carrier recovery
1: Input: received frame $r[n]$, half-symbol length L, pilot indices, true frame layout for metric evaluation.
2: Compute the sliding autocorrelation $M(d)$ and energy $P(d)$ over the search range by cumulative sums, equation (1).
3: Form the normalized timing metric $\Lambda(d)$ of equation (2) and take $\hat{d}$ as the position of its maximum.
4: Estimate the coarse offset from the autocorrelation angle at $\hat{d}$, equation (3), and compensate with the opposite phase ramp.
5: Estimate the residual offset from the pilot phase drift between two consecutive data symbols, equation (4).
6: Compensate the received frame with the total estimate and demodulate; evaluate the genie, uncompensated, and recovered cases.
7: Output: timing error, coarse and total carrier-offset errors, and BER/EVM of the three demodulation cases.
Implementation: generate_preamble.m · schmidl_cox_metric.m · estimate_coarse_cfo.m · estimate_fine_cfo.m · apply_cfo.m · demod_ofdm_symbol.m

The procedure operates as follows. It starts from the received frame, whose preamble was built on the even frequency bins so that its two time-domain halves repeat. The autocorrelation and energy sums of equation (1) are first evaluated over the whole search range with cumulative sums, the normalized metric of equation (2) is formed, and its maximum yields the timing decision. The coarse carrier estimate is then read from the autocorrelation phase at the detected timing according to equation (3), the frame is counter-rotated, and the fine stage measures the remaining rotation between the pilot observations of two consecutive data symbols according to equation (4). The procedure concludes by compensating the frame with the combined estimate and demodulating the three evaluation cases, from which the timing, carrier, and error-rate statistics of Section 6 are accumulated.

3.2 Least-squares and Wiener MMSE channel estimation

Every transmit antenna owns a non-overlapping comb of pilot subcarriers assigned modulo the antenna count, so the pilot observed at any subcarrier originates from exactly one antenna and the per-antenna channels are estimated without pilot contamination [4]. With unit pilots r(p) = 1, the least-squares estimate at pilot subcarrier p of the antenna pair under consideration is the received observation itself,

$$\hat{H}\_LS(p) = Y(p) / r(p) = H(p) + N(p)\,, \tag{5}$$

an unbiased estimate whose error is purely scaled noise, and ls_pilot_estimate.m extends it to the full grid by spline interpolation along frequency, separately per transmit-receive antenna pair. Its total error therefore contains two components, the pilot noise passed through unchanged and the interpolation error across the unobserved subcarriers; the first falls with SNR while the second does not, and their crossover produces the estimation floor measured in Section 6.3.

The Wiener MMSE estimator instead exploits the channel statistics [2], [3]. The frequency-domain correlation between subcarriers k and l depends only on the delay-power profile,

$$R(k,l) = \Sigma_i\, P_i \exp(\, -j\, 2\pi\, \tau_i\, (k-l) / N\,)\,, \tag{6}$$

where P_i and τ_i denote the tap powers and delays of Table 1, and the filter mapping the pilot observations to the full grid combines the all-to-pilot and pilot-to-pilot blocks of that correlation matrix with the noise variance,

$$W = R\_FP\,(\, R\_PP + \sigma^2 I\,)^{-1}. \tag{7}$$

The correlation blocks are precomputed once per configuration by compute_wiener_matrices.m, only the noise-dependent inversion is rebuilt per SNR point, and wiener_mmse_estimate.m applies the filter per antenna pair. Estimation quality is measured as the mean-square error against the true pilot-time channel, averaged over antenna pairs, subcarriers, and slots. Figure 2 presents both estimators as one workflow, whose left branch is the interpolating least-squares path and whose right branch is the statistics-aware Wiener path.

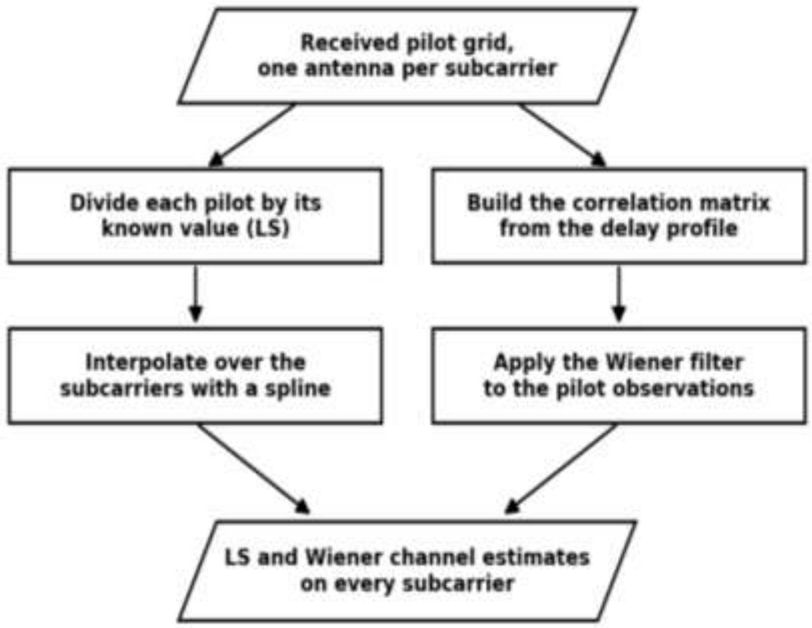

Figure 2. Channel-estimation workflow: least-squares pilot division with spline interpolation on the left branch and delay-profile statistics with Wiener filtering on the right branch, equations (5)–(7).

Algorithm 2: Comb-pilot channel estimation, LS and Wiener MMSE
1: Input: received pilot grid Y, per-antenna pilot index combs, delay-power profile (P_i, τ_i), noise variance σ^2.
2: Extract, per transmit antenna, the pilot observations of its comb; the LS estimate at every pilot equals the observation, equation (5).
3: Interpolate the LS estimates to all subcarriers with a spline, separately per transmit-receive antenna pair.
4: Precompute the correlation blocks R_PP and R_FP from the delay-power profile, equation (6), once per configuration.
5: Build the Wiener filter of equation (7) for the current noise variance and apply it to the pilot observations per antenna pair.
6: Output: Ĥ_LS and Ĥ_MMSE over the full grid, and their mean-square errors against the true pilot-time channel.
Implementation: ls_pilot_estimate.m · compute_wiener_matrices.m · wiener_mmse_estimate.m

The procedure operates as follows. It starts from the received pilot symbol, in which every subcarrier carries the pilot of exactly one transmit antenna by the comb construction. For each antenna the pilot observations of its comb are extracted, and because the pilots have unit power, the least-squares estimate at every pilot position is the observation itself, equation (5); a spline then carries these estimates across the unobserved subcarriers for every antenna pair. In parallel, the correlation blocks of equation (6), computed once from the declared delay-power profile, combine with the current noise variance into the Wiener filter of equation (7), which maps the same pilot observations to the full grid with statistically optimal smoothing. The procedure concludes by delivering both estimates to the detector of Section 3.3 and accumulating their mean-square errors against the true channel.

3.3 Linear detection: zero-forcing and soft-output unbiased MMSE

On subcarrier k the detector observes the power-split MIMO relation

$$y = (H / \sqrt{N\_t})\, x + n\,, \quad n \sim CN(0, \sigma^2 I)\,, \tag{8}$$

where x denotes the N_t×1 vector of unit-power layer symbols, H the N_r×N_t channel matrix of that subcarrier, and σ^2 the per-antenna noise variance fixed by the swept SNR; the $1/\sqrt{N\_t}$ scaling enforces the equal-total-power convention of Section 2 and is applied identically to the true channel and to both channel estimates. The zero-forcing detector inverts the estimated channel by the Moore-Penrose pseudo-inverse,

$$\hat{x}\_ZF = \hat{H}^{+} y\,, \tag{9}$$

removing inter-stream interference exactly when the estimate is exact, at the cost of noise enhancement whenever the channel is poorly conditioned [8]; zf_equalize_mimo.m implements this branch. The MMSE detector regularizes the inversion with the noise variance,

$$W\_eq = (\hat{H}^H \hat{H} + \sigma^2 I)^{-1} \hat{H}^H, \quad \hat{x} = W\_eq\, y\,, \tag{10}$$

converging to zero forcing at high SNR and to a matched filter at low SNR [10]. The raw MMSE output is amplitude-biased toward the origin and its reliability differs across streams, so the soft-output formulation of mmse_equalize_soft.m reads the per-stream gain from the diagonal of the composite response, removes the bias, and forms the per-stream effective noise variance from the same quantity,

$$g_i = [\, W\_eq\, \hat{H}\,]_{ii}\,, \quad \hat{x}_i \leftarrow \hat{x}_i / g_i\,, \quad \sigma^2\_eff{,}i = (1 - g_i) / g_i\,, \tag{11}$$

a model-based per-stream effective-error variance under the assumed linear-MMSE model. Under the matched linear-MMSE model the stream gain provides this quantity directly; when estimated CSI is used, it represents the receiver's reliability estimate rather than the exact realized stream-error variance. Bit log-likelihood ratios are then demapped from the unbiased symbols under the max-log approximation with this per-stream variance as the scaling input,

$$LLR_i(z) \approx (1/\sigma^2\_eff) [min\_\{a\in S_i^{\,0}\} |z-a|^2 - min\_\{a\in S_i^{\,1}\} |z-a|^2] , \tag{12}$$

so that streams with poor post-equalization SINR contribute appropriately weak reliabilities to the decoder [12]. This per-stream calibration is the mechanism by which the soft decoder exploits the reliability spread that a large linear-detected array produces, and it is the direct cause of the growing soft-decision gain measured in Section 7. Figure 3 presents the detection pipeline.

Figure 3. Detection workflow: per-subcarrier observation, the zero-forcing branch, and the soft-output MMSE branch with bias removal, per-stream effective noise variance, and calibrated LLR demapping, equations (8)–(12).

Algorithm 3: Per-subcarrier detection with soft outputs
1: Input: received vector y, scaled channel estimate $\hat{H}/\sqrt{N\_t}$ (LS for the ZF branch, Wiener for the MMSE branch), noise variance σ^2.
2: Zero-forcing branch: recover $\hat{x}\_ZF$ by the pseudo-inverse of equation (9).
3: MMSE branch: build the regularized weight of equation (10) and form the biased estimate.
4: Read the per-stream gains g_i from the diagonal of $W\_eq\hat{H}$, remove the bias, and form σ^2_eff,i, equation (11).

5: Demap hard bits from the common unbiased MMSE estimate, and generate calibrated LLR values from the same estimate under equation (12) using the corresponding per-stream effective-error variance.

6: Output: hard bit stream, LLR stream, and the per-stream quantities feeding the SINR and EVM metrics.

Implementation: zf_equalize_mimo.m · mmse_equalize_soft.m · qamdemod (bit and approxllr outputs)

The procedure operates as follows. It starts from the per-subcarrier observation of equation (8) and the scaled channel estimate delivered by Algorithm 2. The zero-forcing branch inverts the estimate directly by equation (9) and characterizes the cost of exact interference removal. The MMSE branch first forms the regularized weight of equation (10), then converts its biased output into a calibrated soft observation: the per-stream gain read from the composite response removes the amplitude bias, and the same gain yields the model-based per-stream effective-error variance, equation (11). The procedure concludes with two demappings of the common unbiased symbol stream, hard bits for the hard-decision decoder and variance-scaled log-likelihood ratios, equation (12), for the soft-decision decoder, so that the two decoder branches differ only in the use of hard or soft reliability information.

3.4 Threshold-Based Adaptive Modulation and Fixed-Rate Dual-Decoder Channel Coding

Adaptive modulation selects the constellation from the operating SNR through the fixed thresholds of Table 1; the code rate is fixed at 1/2 throughout, so the executed scheme is adaptive modulation with a fixed-rate code rather than complete adaptive modulation and coding. Accordingly, every raw BER curve of this paper exhibits a characteristic sawtooth, falling within each modulation region and jumping upward at each switch to a denser constellation; the sawtooth is the visible signature of link adaptation acting on the chain [9]. Channel coding uses the rate-1/2 constraint-length-7 convolutional code with generators [133 171]. The common gain-corrected unbiased MMSE symbol stream of Section 3.3 feeds two Viterbi decoders [11]: the hard-decision decoder demaps the unbiased symbols to bits and applies the Hamming metric, while the soft-decision decoder operates on the calibrated log-likelihood ratios of equation (12) with an unquantized metric [12]. Because both branches consume the identical unbiased symbols from the same channel realizations, the measured separation between their coded error rates is attributable to the use of hard or soft reliability information. The coding gain is read as the horizontal SNR distance between the two coded curves at a fixed target error rate,

$$\Delta SNR(target) = SNR\_hard(target) - SNR\_soft(target)\,, \tag{13}$$

with each crossing interpolated on the logarithmic BER axis within the highest-modulation region where both curves are monotonic, implemented by interp_snr_at_ber.m; the readout reports a crossing only when the curve genuinely crosses the target inside the measured sweep. Figure 4 presents the link-adaptation and dual-decoding workflow.

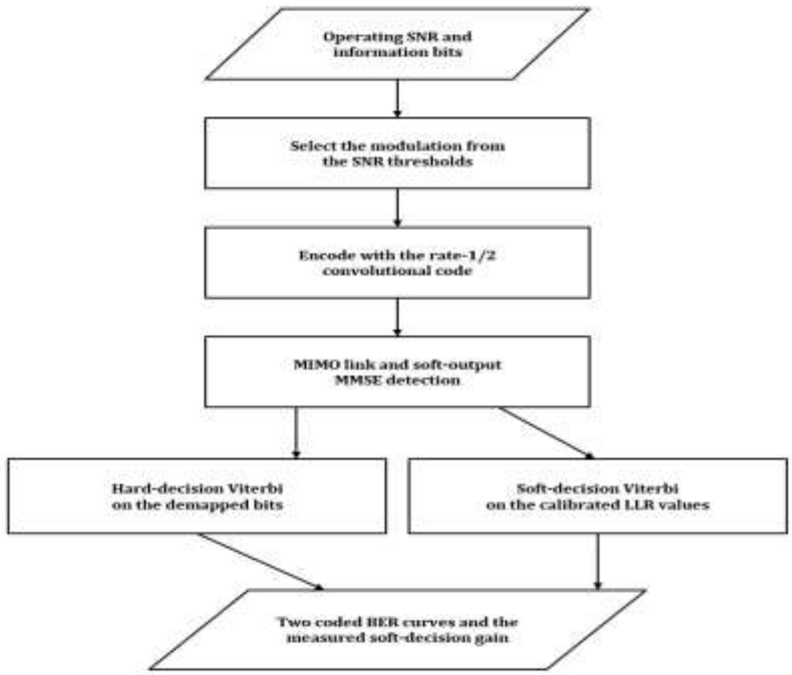

Figure 4. Link-adaptation and coding workflow: threshold-based modulation selection, convolutional encoding, the shared MIMO detection stage, and the two Viterbi decoders whose separation defines the measured coding gain, equations (12)–(13).

Algorithm 4: Threshold-based adaptive modulation with fixed-rate hard- and soft-decision decoding
1: Input: configured operating SNR, information bits, modulation-selection thresholds, and the fixed rate-1/2 convolutional-code trellis. 2: Select the modulation order from the SNR thresholds and size the slot payload accordingly. 3: Encode the information bits with the rate-1/2 convolutional code and map to unit-power QAM symbols. 4: Transmit through the MIMO link and detect with the soft-output MMSE stage of Algorithm 3. 5: Decode the hard bit stream with hard-decision Viterbi; decode the LLR stream with soft-decision (unquantized) Viterbi. 6: Accumulate both coded BER values; at readout, interpolate the target-BER crossings of both curves on the logarithmic axis, equation (13), and report their separation as the measured coding gain. 7: Output: hard and soft coded BER versus SNR and the measured soft-decision gain of the top-modulation region.

Implementation: amc_select_modulation.m · convenc · vitdec ('hard' and 'unquant') · interp_snr_at_ber.m

The procedure operates as follows. It starts by selecting the modulation order from the operating SNR and sizing the coded payload of the slot to it. The information bits pass through the convolutional encoder and onto the MIMO link, and the detection stage of Algorithm 3 returns, from the same equalizer output, a directly demapped hard bit stream and a calibrated LLR stream. The two Viterbi decoders then run side by side, and their coded error rates accumulate over the Monte Carlo slots. The procedure concludes at readout, where the crossing of each coded curve with the target error rate is interpolated on the logarithmic axis within the top-modulation region, equation (13), and the horizontal separation of the two crossings is reported as the measured soft-decision coding gain; when a curve does not cross the target inside the measured sweep, the readout reports that fact instead of extrapolating.

3.5 Performance metrics

Every metric reported in Sections 6 and 7 is defined here. Link reliability is measured by the bit error rate over the accumulated bits; modulation accuracy by the RMS error vector magnitude between the equalized and transmitted symbols; detection quality by the reference-based post-equalization SINR; estimation accuracy by the mean-square error of the channel estimate against the true pilot-time channel; and the information-theoretic reference by the per-subcarrier ergodic capacity of Telatar under equal power allocation [5],

$$BER = N\_err/N\_bit\,,\;\; EVM = \sqrt{\Sigma|\hat{x}-x|^2 / \Sigma|x|^2}\,,\;\; SINR = \Sigma|x|^2 / \Sigma|\hat{x}-x|^2\,,\;\; C = (1/N)\, \Sigma_k\, \Sigma_i\, \log_2(\,1+\rho\,\sigma_i^{\,2}(k)\,)\,, \tag{14}$$

computed by compute_evm.m, compute_sinr.m, and compute_mimo_capacity.m from identical symbol definitions in both configurations. The hard-decision coded goodput is reported as a goodput-style figure, the surviving information rate scaled by one minus the hard-decision coded BER, and spectral efficiency is that goodput normalized per subcarrier; both are declared hard-decision proxies rather than measured block-delivery rates, a scope boundary revisited in Section 8. The capacity of equation (14) is reported as a reference bound and never interpreted as achieved coded throughput.

3.6 End-to-end procedure, workflow, and traceability

Algorithm 5 states the complete end-to-end procedure of one Monte Carlo slot in execution order, and Figure 5 presents the same procedure as a workflow diagram from the configuration input to the stored result file. Table 2 completes the correspondence by mapping every algorithm to its derivation, its implementing MATLAB functions, and the result figures it produces, so each reported curve is traceable to the mathematics of this section and to a specific source file.

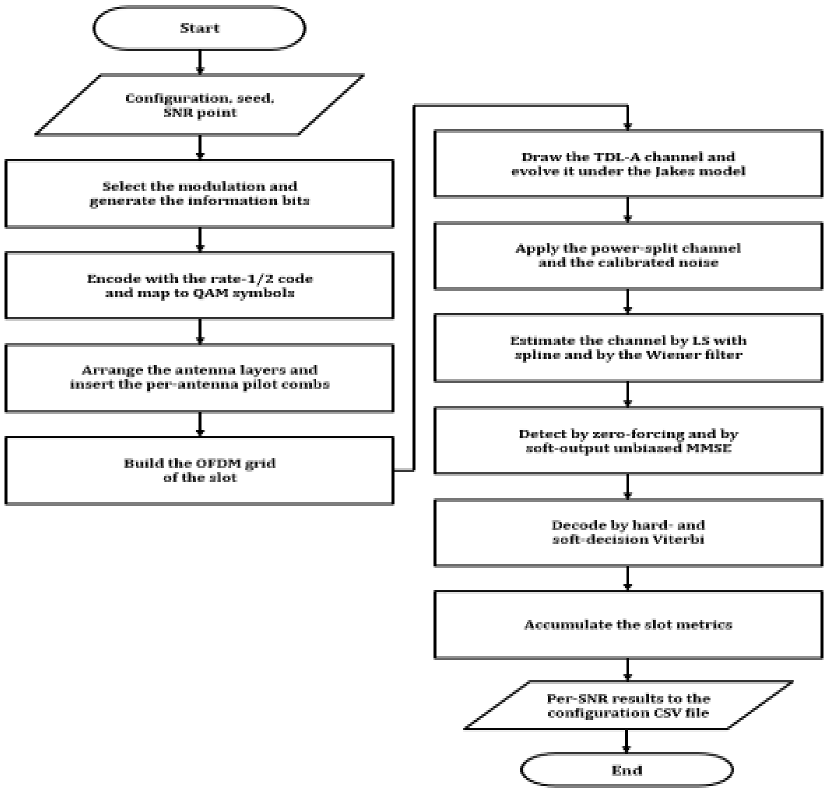

Figure 5. End-to-end workflow of one Monte Carlo slot, from the configuration input through the transmitter, channel, receiver, and decoding stages to the per-configuration result file; the implementing MATLAB functions of every step are listed in Table 2.

Algorithm 5: End-to-end link-level simulation (one Monte Carlo slot)
1: Select the modulation order from the operating SNR (Algorithm 4) and generate the information bits.
2: Encode with the rate-1/2 convolutional code and map to unit-power QAM symbols.
3: Arrange the symbols into the antenna layers and insert the orthogonal per-antenna pilot combs into the OFDM grid.
4: Draw one TDL-A channel realization and evolve it per OFDM symbol under the simplified common-Doppler phase-evolution model.
5: Apply the power-split channel and the fixed-variance receiver noise, equation (8).
6: Estimate the channel from the pilot symbol by LS with spline interpolation and by the Wiener filter (Algorithm 2), equations (5)–(7).
7: Detect every data resource element with zero-forcing and with the soft-output MMSE stage (Algorithm 3), equations (9)–(12).
8: Decode the coded stream with hard- and soft-decision Viterbi (Algorithm 4).

9: Accumulate the metrics of equation (14) and, at the end of the sweep, write the aggregated results to the per-configuration CSV file.

Implementation: main_phy_simulation.m (8×8, Parts A and B) · main_phy_simulation_16x16.m (16×16) · run_size_comparison.m (overlays and crossings)

The slot proceeds as follows. It starts with the modulation selection and the generation of one information payload, which is encoded and mapped to unit-power symbols. The symbols are arranged across the transmit antennas and placed, together with the per-antenna pilot combs, into the OFDM grid. One channel realization is drawn and evolved symbol by symbol under the simplified common-Doppler phase-evolution model, the power-split channel and the calibrated noise are applied, and the receiver estimates the channel from the pilot symbol by both estimators, detects every data resource element by both detectors, and decodes the coded stream by both decoders. The slot concludes with the metric accumulation, and the sweep concludes by writing the aggregated result table to disk, from which every result figure and every quoted numerical result of Sections 6 and 7 is generated.

Table 2. Traceability of the algorithms: derivation, implementing MATLAB functions, and the result figures each produces.

Algorithm	**Equations**	**MATLAB implementation**	**Result figures**
1. Timing and carrier recovery	(1)–(4)	generate_preamble.m, schmidl_cox_metric.m, estimate_coarse_cfo.m, estimate_fine_cfo.m, apply_cfo.m, demod_ofdm_symbol.m	Figures 6–10
2. LS and Wiener estimation	(5)–(7)	ls_pilot_estimate.m, compute_wiener_matrices.m, wiener_mmse_estimate.m	Figures 12, 20, 27
3. ZF and soft-output MMSE	(8)–(12)	zf_equalize_mimo.m, mmse_equalize_soft.m	Figures 11, 13, 14, 19, 21, 22, 28
4. Threshold-based adaptive modulation and dual decoding	(12)–(13)	amc_select_modulation.m, convenc, vitdec, interp_snr_at_ber.m	Figures 16–18, 24, 25, 29
5. Performance metrics	(14)	compute_evm.m, compute_sinr.m, compute_mimo_capacity.m	Figures 13–15, 21–23, 26, 28
6. End-to-end slot and provenance	(1)–(14)	main_phy_simulation.m, main_phy_simulation_16x16.m, run_size_comparison.m	Figures 15, 23, 26–29

The first column of Table 2 names the numbered procedures of this section, the second column lists the equations that each realizes, the third column names the MATLAB functions that implement it, and the final column points to the executed result figures the procedure produces. The table serves the same purpose as the workflow diagram in tabular form: every reported curve of this paper can be walked back through its figure to the procedure, the equations, and the source file that generated it.

4. Simulator Architecture and Result Provenance

Section 4 describes the structure of the simulation package and the provenance chain through which every reported number is traceable to a stored file. Section 4.1 states the package structure and the division of responsibility between the scripts, and Section 4.2 states the result-provenance rules.

4.1 Package structure

The package contains two coordinated studies, an OFDM synchronization study and an 8×8/16×16 MIMO-OFDM link-level study, executed by the same script set; the synchronization estimates characterize the acquisition algorithms and do not feed the MIMO demodulation, whose evaluation uses its own pilot processing. The package consists of twenty-three MATLAB files: two configuration main scripts, one comparison script, one standalone LLR sign-convention unit test, and nineteen single-purpose function files, one file per algorithmic responsibility. The 8×8 main script executes the synchronization study and the 8×8 MIMO study; the 16×16 main script executes the MIMO study only, because synchronization is a single-antenna function whose results are independent of the array size, and repeating it would duplicate identical numbers. Both main scripts run under the fixed seed of Table 1 and export their aggregated per-SNR results as CSV tables, results_8x8.csv and results_16x16.csv, with one named column per metric.

4.2 Result provenance

Provenance is enforced by the consumption path of those files. The comparison script loads the two tables, refuses to run when either file is missing, verifies that the two SNR grids are identical before drawing any overlay, and computes the target-BER crossings of Section 7 from the stored columns by logarithmic interpolation rather than from re-typed values. The guarded readout of equation (13) belongs to the same discipline: when a stored curve does not cross the requested target inside the measured sweep, the script reports that fact, and Section 7 quotes the refusal as a result. Every figure of Sections 6 and 7 is therefore regenerable from the two CSV files and the scripts alone.

5. Verification Methodology and Defect Record

Section 5 documents the verification methodology and the record of the four defects it exposed: Section 5.1 the corrupted genie reference, Section 5.2 the missing FFT processing gain, Section 5.3 the preamble subcarrier parity, and Section 5.4 the inconsistent MMSE decoder streams. Verification followed an execution-first discipline: the chain was run, every metric was compared against its theoretically required behavior, and every inconsistency was treated as a defect to be isolated with an independent numerical experiment before the code was changed. Each defect is documented with its observed symptom, its diagnosis, the correction, and the post-fix confirmation, because the record of how correctness was established is part of the result.

5.1 Corrupted genie synchronization reference

The genie reference of the synchronization study was originally constructed by applying the negative true carrier offset to the channel output before noise, a signal that carried no offset in the first place, so the compensation itself injected a spurious quarter-subcarrier offset and inter-carrier interference into the supposedly ideal case. The symptom was a perfect-synchronization curve that performed implausibly poorly at high SNR. The correction compensates the actually received signal with the true offset, the proper definition of the genie case, and after the fix the reference forms the lower envelope of Figure 7 as ideal synchronization must.

5.2 FFT processing gain in the noise calibration

The synchronization part originally set the time-domain noise variance to the reciprocal of the target SNR, overlooking that the OFDM waveform, generated by an inverse FFT of unit-power symbols without rescaling, carries average time-domain power of 1/N. The demodulated per-subcarrier SNR was therefore below the labeled value by the FFT length, a factor of 24 dB, which explained an entire family of anomalies at once: bit error rates pinned near one half, error vector magnitudes in the hundreds of percent, and timing errors above one hundred samples at nominally high SNR. The relationship was confirmed by an independent numerical experiment before the change, measuring the demodulated per-subcarrier SNR under both scalings against a 20 dB target and reproducing the 24 dB discrepancy exactly. The corrected variance divides by the FFT length, and the rerun transformed every synchronization metric into its textbook form.

5.3 Preamble subcarrier parity

The most instructive defect was an indexing subtlety. The preamble generator placed its training symbols with the MATLAB index pattern 2:2:N, which under one-based indexing selects the odd frequency bins in the zero-based sense; energy on odd bins makes the two time-domain half-symbols anti-identical rather than identical, negating the autocorrelation of equation (1). The magnitude-based timing metric of equation (2) is blind to the sign and continued to produce a healthy-looking plateau, which concealed the defect, but the coarse estimator of equation (3) reads the correlation phase, and the extra half-turn displaced its estimate by exactly one subcarrier spacing. The symptom was correspondingly precise: a carrier-offset RMSE pinned at 1.0 independently of SNR and a recovered bit error rate of one half, since compensation with an estimate one full subcarrier away shifts every subcarrier by one bin. A standalone numerical reconstruction confirmed both the anti-identical halves and a coarse estimate of −0.75 for a true offset of +0.25 under the defective indexing, and the identical construction with the pattern 1:2:N restored identical halves and an exact estimate. After the one-line correction, the carrier-offset RMSE fell monotonically from 0.16 at 0 dB to 0.001 at 30 dB and the recovered curve joined the genie reference.

5.4 Inconsistent MMSE inputs in the hard/soft decoder comparison

The original hard-decision branch demapped bits from the amplitude-biased MMSE output, while the soft-decision branch generated log-likelihood ratios from the gain-corrected unbiased output. Consequently, the measured separation combined the effect of MMSE gain correction with the effect of soft reliability information. The receiver was corrected so that both branches use the same gain-corrected unbiased MMSE symbols, and the same unbiased stream now also feeds the uncoded MMSE EVM and SINR metrics. The hard branch converts these symbols to binary decisions, whereas the soft branch converts the same symbols to per-stream variance-scaled log-likelihood ratios. Both array configurations were rerun, and every decoder-dependent result was regenerated.

Table 3. Defects found through execution, with observed symptom and resolution.

Defect	Observed symptom	Independent check	Resolution
Genie reference compensated the wrong signal	Perfect-sync BER implausibly high	Definition audit of the genie case	Compensate the received signal with the true offset
Time-domain noise variance missing 1/N	BER ≈ 0.5, EVM in hundreds of percent, timing RMSE > 100 samples at high SNR	Two-scaling SNR measurement reproducing the 24 dB gap	Scale the noise variance by the FFT length
Preamble energy on odd frequency bins	CFO RMSE pinned at exactly 1.0 at all SNR; recovered BER ≈ 0.5; timing metric deceptively healthy	Standalone reconstruction: anti-identical halves, estimate −0.75 for true +0.25	Even-bin placement, index pattern 1:2:N
Different MMSE streams for hard and soft decoding	Decoder gap included the MMSE gain-correction difference	Compare both branches from one common unbiased stream	Hard and soft demappers now use the same gain-corrected symbols

The first column of Table 3 names each defect, the second states the symptom under which it presented in the executed results, the third states the independent numerical experiment that localized it before any code was modified, and the final column states the correction. The table makes the discipline concrete: none of the four defects was fixed on suspicion, each was reproduced in isolation first, and each fix was accepted only after the rerun restored the theoretically required behavior. The parity defect additionally illustrates why magnitude-level health checks are insufficient, a metric that looked correct concealed a phase error that destroyed the chain downstream.

6. Executed Results

Section 6 reports the executed results of both configurations: Section 6.1 the synchronization performance of Algorithm 1, Section 6.2 the MIMO chain performance of Algorithms 2 to 4 at 8×8 with the headline summary of Table 4, and Section 6.3 the MIMO chain performance at 16×16 with the headline summary of Table 5.

6.1 Synchronization performance

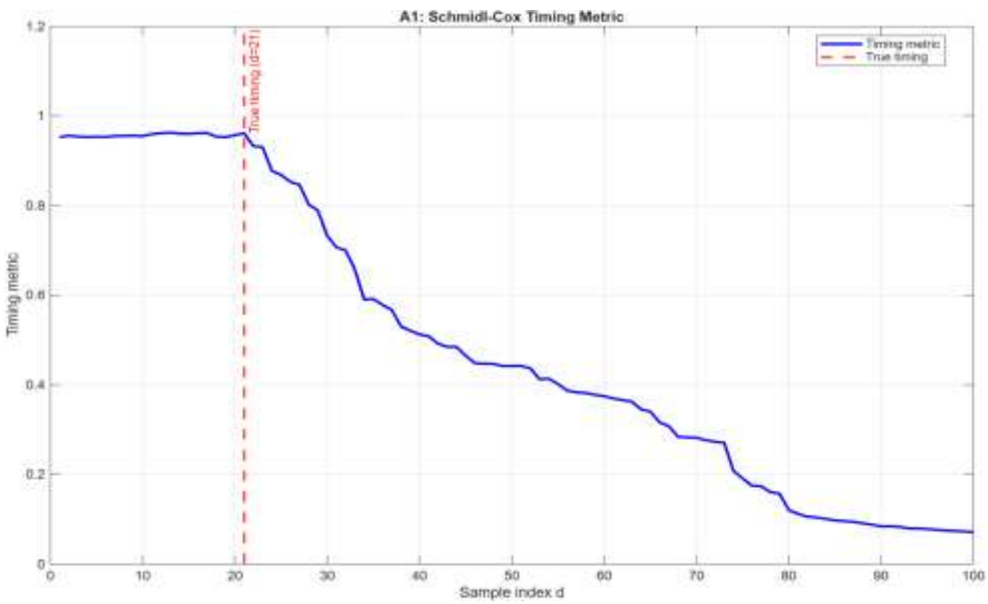

Figure 6. Schmidl-Cox timing metric of equation (2) for one channel realization at 20 dB SNR, with the true timing position marked (schmidl_cox_metric.m).

Figure 6 shows that the timing metric holds a plateau close to unity across the cyclic-prefix region, reaches its maximum within the valid timing plateau near sample 21, and rolls off smoothly beyond the plateau. The flat top is the documented property of the original Schmidl-Cox construction: every window position inside the cyclic prefix straddles two identical half-symbols equally well, so the maximum is genuinely ambiguous across that span. The ambiguity is benign here because any timing choice on the plateau keeps the FFT window within the cyclic prefix.

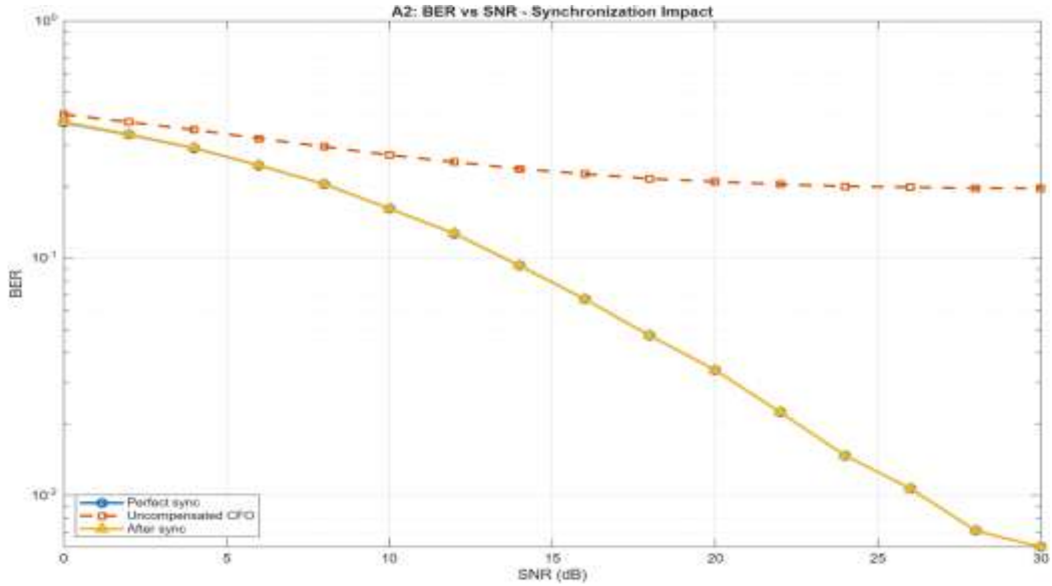

Figure 7. BER against SNR for the three synchronization cases of Algorithm 1: genie reference, uncompensated carrier offset, and the recovered chain.

Figure 7 shows the recovered curve lying on top of the genie reference at every SNR point, with the plotted markers overlapping, which demonstrates that the full timing-plus-carrier-recovery chain returns essentially all of the performance that ideal synchronization would give; at 30 dB the two error rates agree to four significant figures, 6.060×10^{-3} against 6.059×10^{-3}. The uncompensated curve tells the opposite story: it flattens onto an inter-carrier-interference floor near a bit error rate of two in ten, and no amount of additional SNR moves it, because a quarter-subcarrier offset destroys subcarrier orthogonality independently of the noise level.

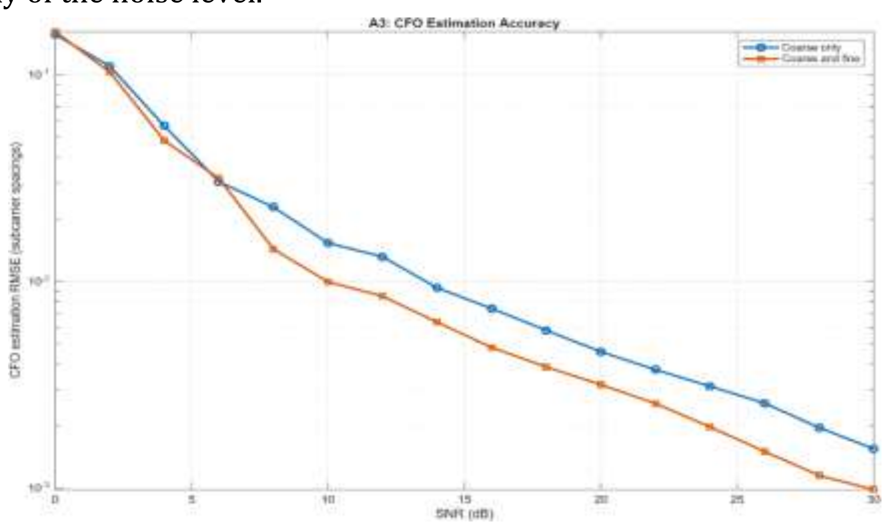

Figure 8. Carrier-offset estimation RMSE for the coarse estimator of equation (3) alone and for the coarse-plus-fine combination of equations (3)–(4).

Figure 8 shows both estimators descending monotonically on the logarithmic axis from roughly one tenth of a subcarrier spacing at 0 dB to the order of one thousandth at 30 dB, the signature of noise-limited estimation. The fine correction pulls below the coarse estimate from about 6 dB upward and roughly halves the residual error at high SNR. Below that point the two curves touch and briefly cross, which is expected: at very low SNR the pilot observations feeding the fine stage are themselves unreliable, so the correction adds as much noise as it removes until the pilots become trustworthy.

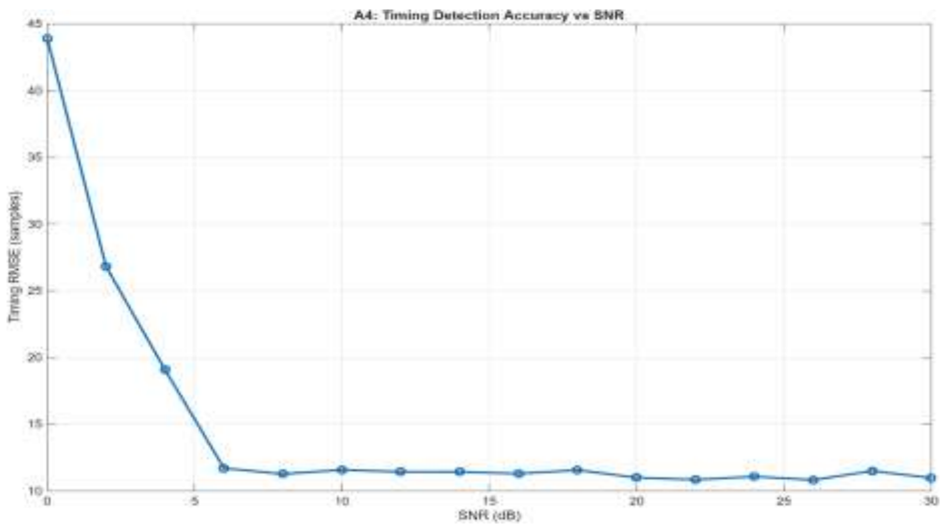

Figure 9. Timing detection RMSE against SNR.

Figure 9 shows that the timing error falls steeply from approximately 44 samples at 0 dB and settles at approximately 11 samples from 6 dB onward. The floor is not residual noise but the plateau ambiguity visible in Figure 6: with a near-flat metric across the cyclic-prefix span, the position of the maximum wanders within that span, and the root-mean-square deviation of a roughly uniform choice over it is of the order of the observed 11 samples. Because the wander stays inside the cyclic prefix it carries no bit-error consequence, and sharpened-metric preamble variants exist precisely to trade this plateau for a peak when sample-accurate timing is required.

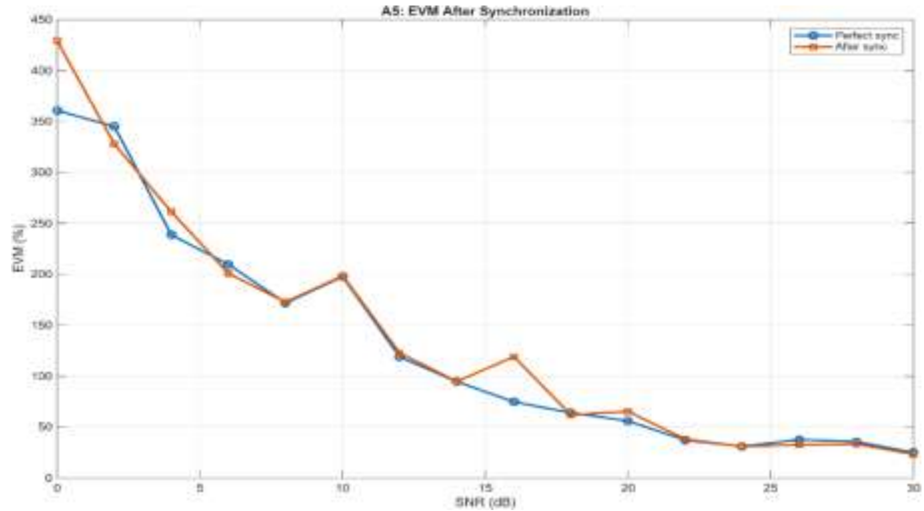

Figure 10. Error vector magnitude for the genie reference and the recovered chain.

Figure 10 presents the two EVM curves tracking each other closely across the sweep, confirming at the modulation-quality level what Figure 7 established at the bit level. The absolute values are large at low SNR because the metric is computed on raw zero-forcing-equalized symbols in Rayleigh fading, where deeply faded subcarriers dominate the root-mean-square through near-zero channel division; this is a property of zero forcing in fading rather than a synchronization effect, and it shrinks steadily as SNR grows.

6.2 MIMO chain performance at 8×8

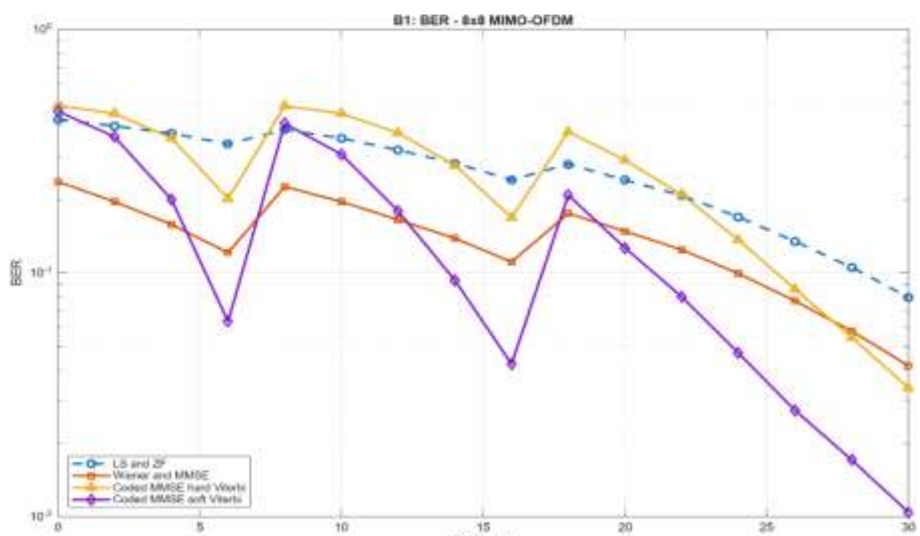

Figure 11. Uncoded and coded BER of the 8×8 chain: LS with zero-forcing, Wiener with unbiased MMSE, and the coded MMSE stream under hard- and soft-decision Viterbi.

Figure 11 shows three orderings holding simultaneously across the sweep. The Wiener-plus-MMSE uncoded curve sits well below the LS-plus-zero-forcing curve at every point, quantifying the joint value of statistical estimation and regularized detection. Both coded curves display the adaptive-modulation sawtooth of Section 3.4, with upward jumps at the 8 dB and 18 dB constellation switches and steep descents inside each region. And the soft-decision curve lies below the hard-decision curve everywhere, reaching one percent at 30 dB, with the hard-decision endpoint at 3.2×10^{-2}.

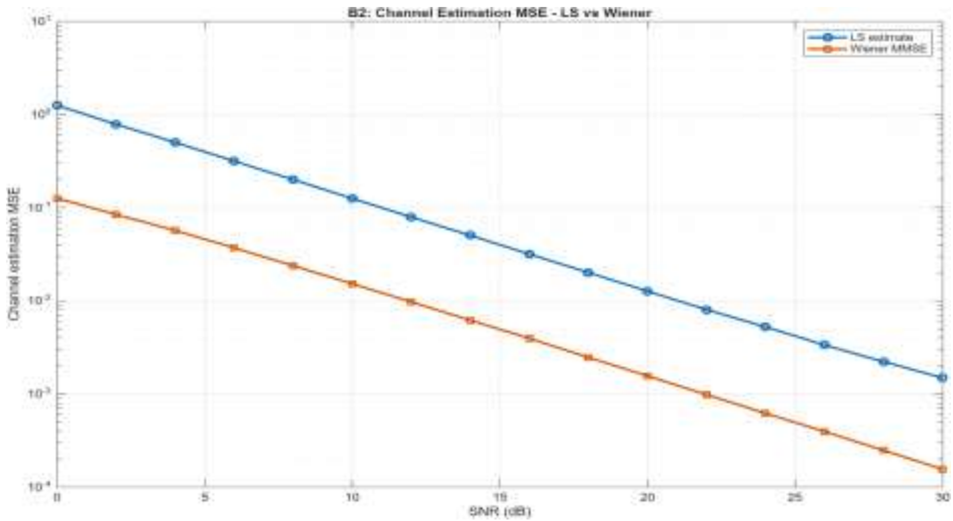

Figure 12. Channel-estimation mean-square error for the LS and Wiener estimators at 8×8, equations (5)–(7).

Figure 12 shows both estimators descending on clean parallel slopes, with the Wiener estimator roughly one decade below least squares across the whole sweep, ending near 1.5×10^{-4} against 1.5×10^{-3} at 30 dB. With 32 pilots per antenna the spline interpolation of the LS estimator is still adequate for this delay spread, so its error remains noise-dominated and keeps falling; the corresponding curve at the doubled array behaves very differently, as Section 6.3 shows.

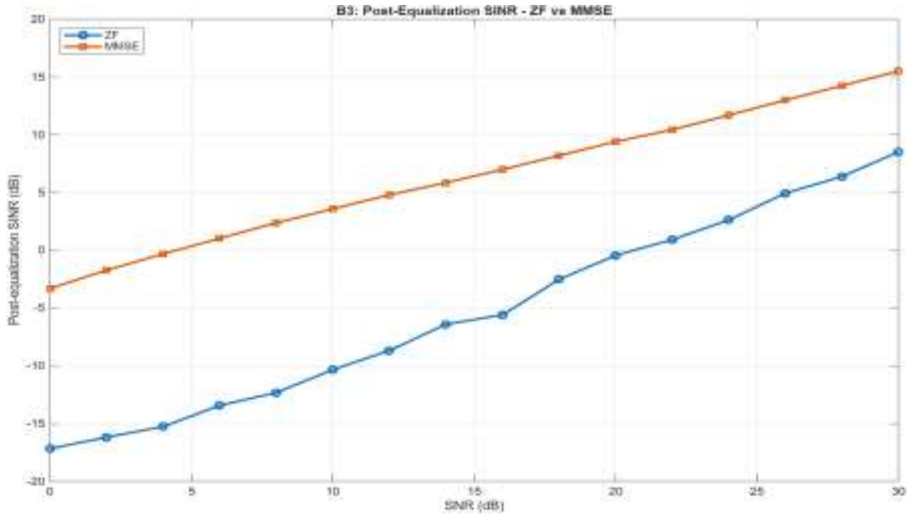

Figure 13. Post-equalization SINR for zero-forcing and unbiased MMSE detection at 8×8.

Figure 13 shows the MMSE output SINR rising steadily with input SNR from about −3 dB at 0 dB, crossing into positive values near 5 dB and reaching 15.5 dB at 30 dB, while zero forcing starts seventeen decibels below zero and only crosses into positive territory near 20 dB input SNR. The vertical distance between the curves is the noise-enhancement penalty of exact interference inversion on an 8×8 fading channel, largest at low SNR exactly where regularization matters most.

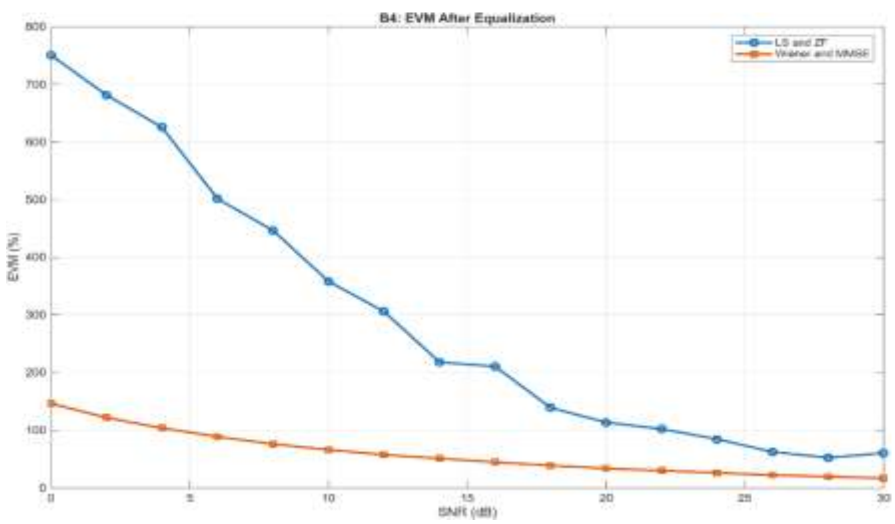

Figure 14. Error vector magnitude after equalization for the two estimator-detector chains at 8×8.

Figure 14 repeats the picture of the SINR curves in modulation-quality terms: the Wiener-plus-MMSE EVM begins near one hundred fifty percent at 0 dB, falls below one hundred percent by about 5 dB, and descends smoothly to about seventeen percent at 30 dB, while the LS-plus-zero-forcing chain begins above seven hundred percent, dominated by the near-singular subcarriers that zero forcing amplifies, and approaches the MMSE chain only at the top of the sweep.

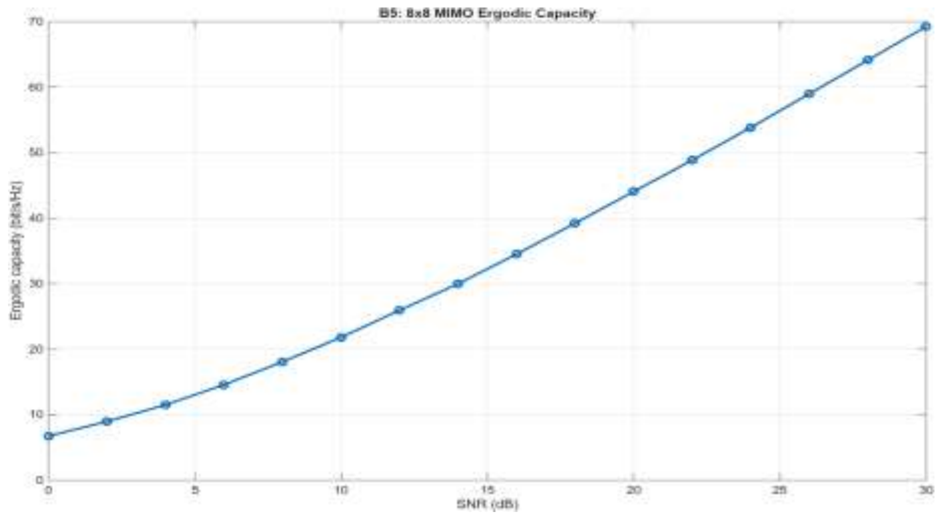

Figure 15. Ergodic capacity of the 8×8 channel, equation (14).

Figure 15 shows the capacity climbing from 6.7 bit/s/Hz at 0 dB to 69.3 bit/s/Hz at 30 dB with the near-linear high-SNR growth expected of a full-rank spatial channel, each of the eight spatial modes contributing its logarithmic share per three decibels.

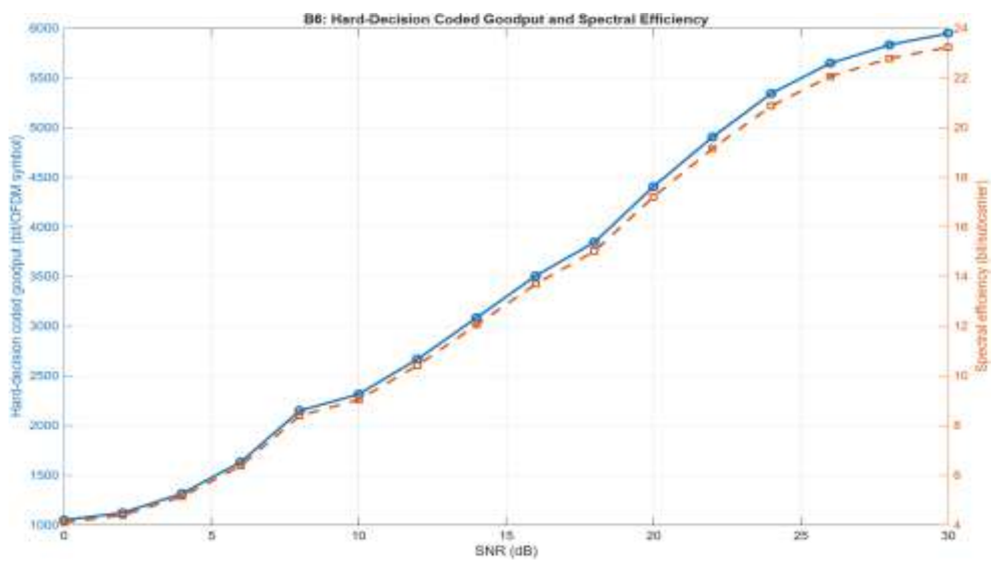

Figure 16. Hard-decision coded goodput and spectral efficiency at 8×8.

Figure 16 shows the hard-decision goodput proxy rising with SNR through each modulation region and stepping upward at the adaptive-modulation switches, reaching roughly 5,900 information bits per OFDM symbol at 30 dB, equivalently about 23 bits per subcarrier, as the 64-QAM region cleans up under the code.

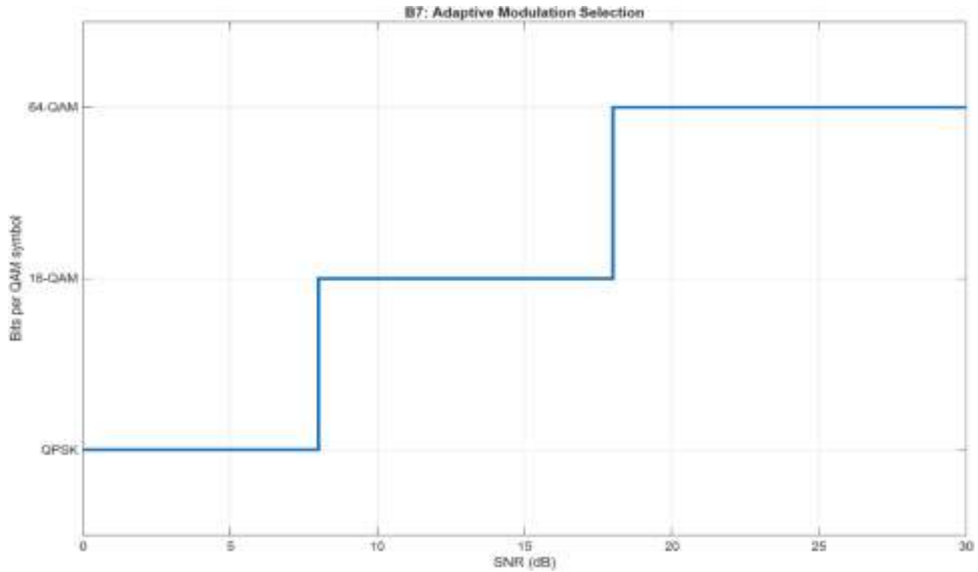

Figure 17. Adaptive modulation selection across the SNR sweep.

Figure 17 presents the selection staircase confirming the configured thresholds of Table 1, QPSK through 6 dB, 16-QAM from 8 through 16 dB, and 64-QAM from 18 dB upward, and provides the reference for reading the sawtooth positions in every BER figure of this paper.

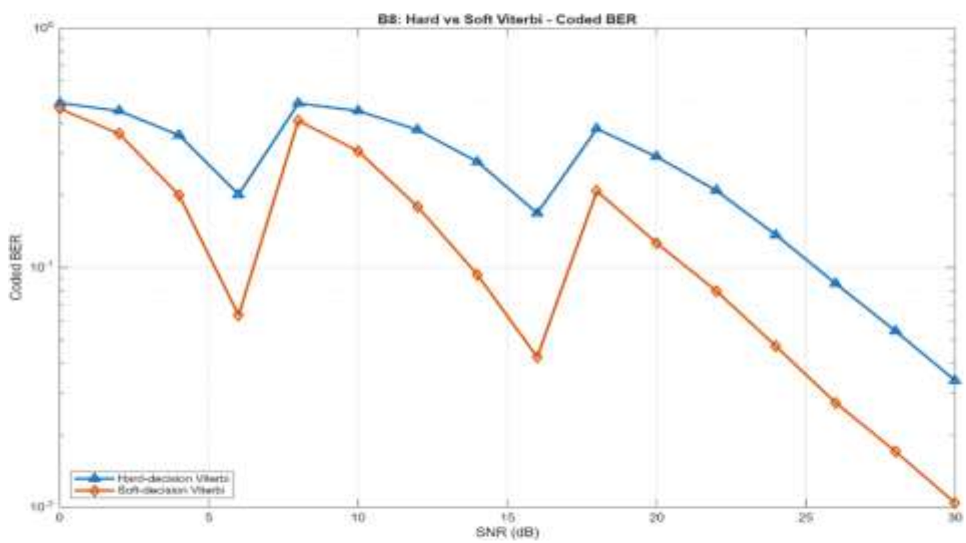

Figure 18. Coded BER under hard- and soft-decision Viterbi decoding at 8×8, the controlled comparison of Algorithm 4.

Figure 18 isolates the decoder comparison and shows the soft decoder below the hard decoder at every SNR point, with both curves sharing sawtooth positions because they decode the same adaptive-modulation stream. Within the 64-QAM region the horizontal separation of equation (13) is interpolated on the logarithmic axis from the stored result table at the 5×10^{-2} and 10^{-1} targets, giving a soft-decision gain of 4.30 dB at 5×10^{-2}, where the hard curve crosses at 28.06 dB and the soft curve at 23.76 dB, and 4.07 dB at 10^{-1}, where the crossings are 25.10 dB and 21.03 dB.

Table 4. Headline measured results of the 8×8 configuration, from results_8x8.csv.

Quantity	Measured value
Recovered vs genie BER at 30 dB	6.060×10^{-3} vs 6.059×10^{-3} (overlapping at all points)
Carrier-offset RMSE at 30 dB	1.0×10^{-3} subcarrier spacings (true offset 0.25)
Timing RMSE floor	≈ 11 samples, plateau-limited, inside the cyclic prefix
Wiener vs LS estimation MSE at 30 dB	1.5×10^{-4} vs 1.5×10^{-3} (≈ 10×)
MMSE vs ZF output SINR at 30 dB	15.5 dB vs 8.5 dB
Ergodic capacity at 30 dB	69.3 bit/s/Hz
Soft-decision separation, 64-QAM region	4.30 dB at 5×10^{-2}, 4.07 dB at 10^{-1}

The first column of Table 4 names the measured quantity and the second states its executed value with the comparison context in parentheses. The rows are ordered along the receiver chain, synchronization first, then estimation, detection, capacity, and coding, so the table reads as a one-page summary of the 8×8 evidence. Every entry is computed from the stored result file rather than read off a figure, and the two coding-gain figures correspond to the two target error rates at which the crossing of equation (13) was evaluated.

6.3 MIMO chain performance at 16×16

The 16×16 run repeats the MIMO chain with the antenna count as the only altered parameter. Synchronization is a single-antenna function executed once in the 8×8 run, and its results of Section 6.1 apply unchanged. The comb pilot pattern divides the same 256-subcarrier grid among sixteen antennas, leaving sixteen pilots per antenna at a spacing of sixteen subcarriers, and the transmit power split, the code, the modulation thresholds, and the Monte Carlo depth are identical to Section 6.2.

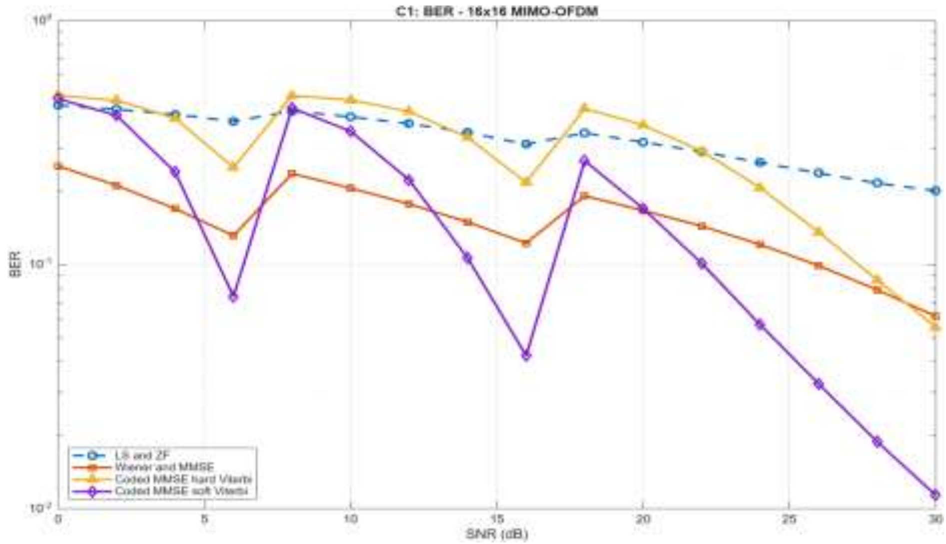

Figure 19. Uncoded and coded BER of the 16×16 chain.

Figure 19 shows the curve family keeping the structure of its 8×8 counterpart, Wiener-plus-MMSE below LS-plus-zero-forcing, the sawtooth at the same switch points, and soft decoding below hard everywhere, but every curve sits higher. The LS-plus-zero-forcing chain degrades the most, ending near 2.0×10^{-1} at 30 dB where the 8×8 chain reached 7.9×10^{-2}, while the soft-decoded chain still reaches 1.1×10^{-2}. The widening spread between naive and statistical processing is the recurring theme of Section 7.

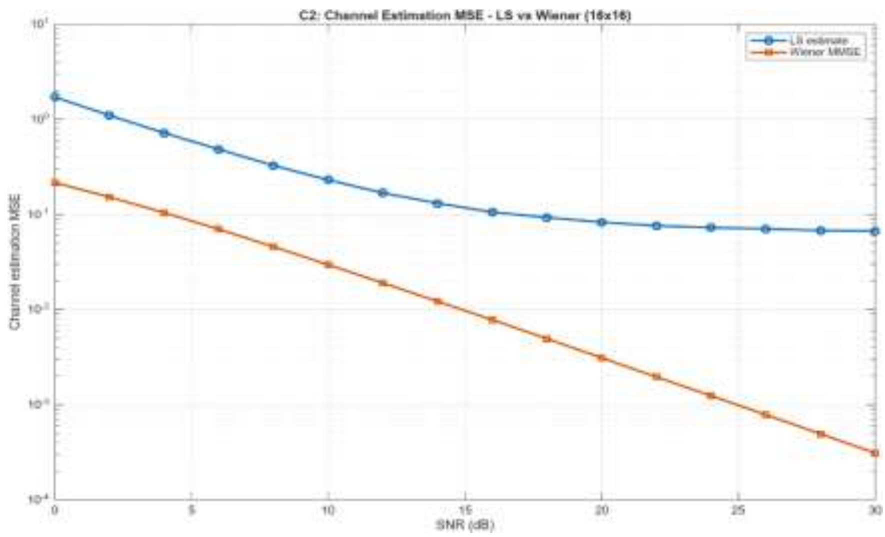

Figure 20. Channel-estimation mean-square error at 16×16.

Figure 20 shows the least-squares curve departing from its noise-limited slope above approximately 18 dB and flattening onto an interpolation-error floor near 7×10^{-2} that additional SNR does not remove. With pilots every sixteen subcarriers, spline interpolation across the channel's frequency selectivity leaves a residual error that dominates once the pilot noise falls below it: the estimator has become interpolation-limited rather than noise-limited, exactly the crossover predicted with equation (5). The Wiener estimator, informed by the true delay-power correlations, keeps its clean slope to 3×10^{-4}, and the ratio between the two estimators at 30 dB has grown to roughly two hundred.

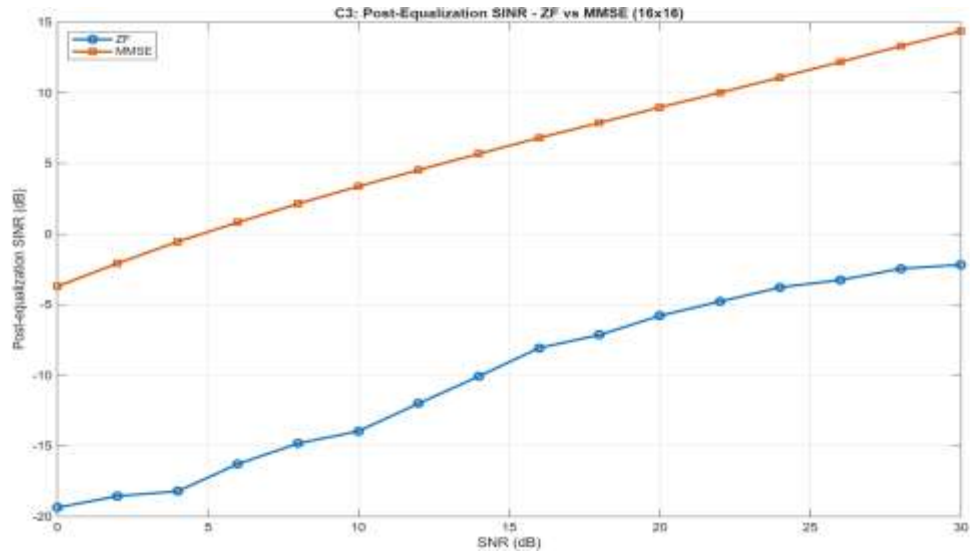

Figure 21. Post-equalization SINR at 16×16.

Figure 21 shows the MMSE output SINR again rising steadily with input SNR from about −3.7 dB at 0 dB to 14.4 dB at 30 dB, about 1.1 dB below its 8×8 counterpart. Zero forcing never recovers: it starts near minus twenty decibels and remains negative across the entire sweep, ending at −2.2 dB, the combined effect of stronger noise enhancement on the larger square inversion and the least-squares estimation floor feeding it a permanently imperfect channel.

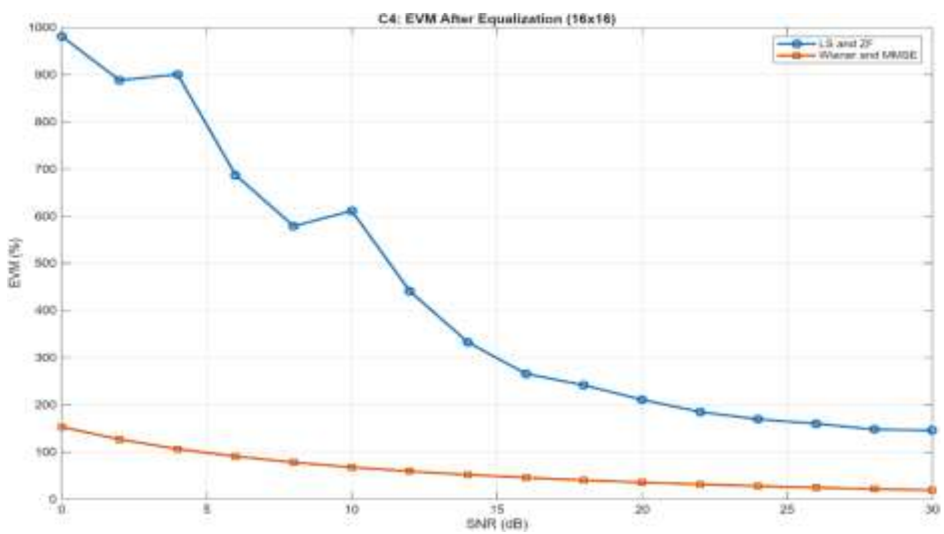

Figure 22. Error vector magnitude after equalization at 16×16.

Figure 22 shows the LS-plus-zero-forcing chain now beginning near one thousand percent and never falls below roughly one hundred forty percent, its floor mirroring the estimation floor of Figure 20, while the Wiener-plus-MMSE EVM begins near one hundred fifty percent at 0 dB and descends smoothly to about nineteen percent at 30 dB.

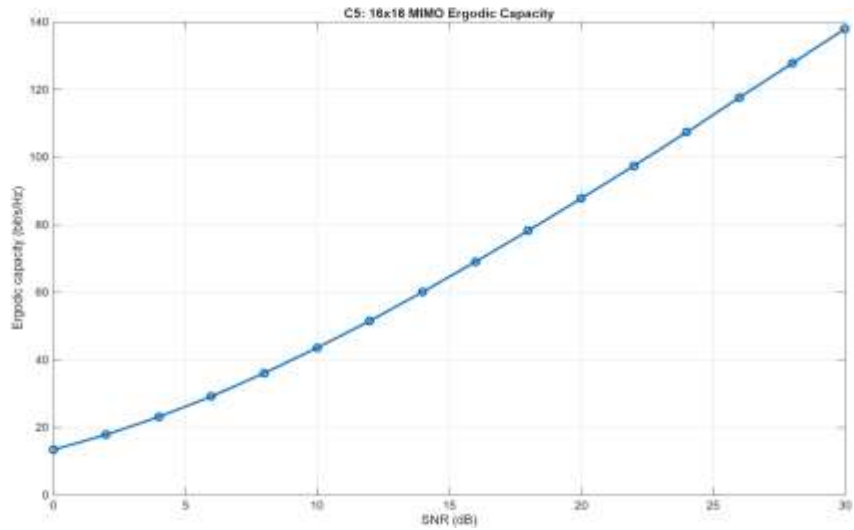

Figure 23. Ergodic capacity of the 16×16 channel.

Figure 23 shows the capacity rising from 13.4 bit/s/Hz at 0 dB to 138.0 bit/s/Hz at 30 dB, twice the 8×8 values across the sweep to within Monte Carlo precision, the direct consequence of doubling the number of spatial modes.

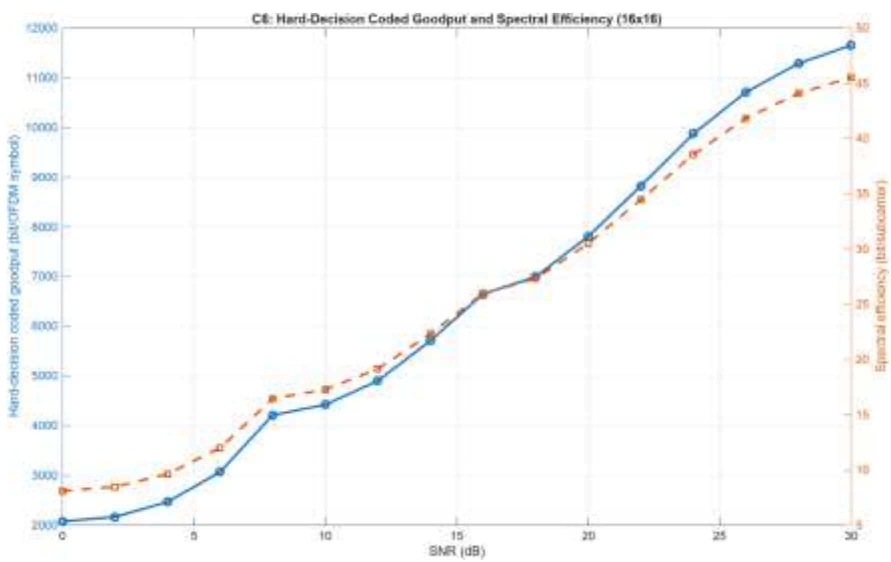

Figure 24. Hard-decision coded goodput and spectral efficiency at 16×16.

Figure 24 shows the peak hard-decision goodput proxy approximately doubling relative to the smaller array, reaching about 11,600 information bits per OFDM symbol, or 45 bits per subcarrier, at 30 dB, slightly below an exact doubling because the residual hard-decision coded error rate at 16×16 is higher.

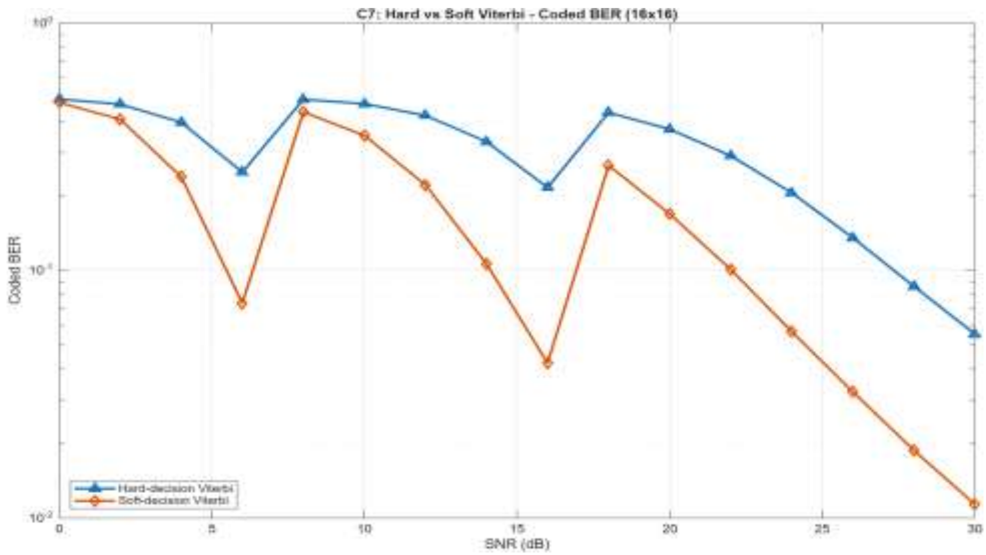

Figure 25. Coded BER under hard- and soft-decision Viterbi at 16×16.

Figure 25 shows the soft decoder maintaining its advantage at the doubled array size, sitting below the hard decoder at every point with a visibly wider gap than at 8×8. The soft curve ends the sweep at 1.1×10^{-2}, nearly identical to its 8×8 endpoint, while the hard-decision endpoint and its displacement relative to the 8×8 curve are 5.2×10^{-2} and roughly two decibels: the 16×16 hard curve ends at 5.2×10^{-2} against 3.2×10^{-2} at 8×8 and crosses a coded BER of one in ten at 27.09 dB against 25.10 dB. Soft decoding absorbs most of the penalty that the larger linear-detected array imposes.

Table 5. Headline measured results of the 16×16 configuration, from results_16x16.csv.

Quantity	Measured value
Ergodic capacity at 30 dB	138.0 bit/s/Hz (2.0× the 8×8 value)
LS estimation floor	≈ 7×10^{-2} above 20 dB (interpolation-limited)
Wiener estimation MSE at 30 dB	3×10^{-4} (≈ 200× below LS)
ZF output SINR at 30 dB	−2.2 dB (never positive across the sweep)
MMSE output SINR at 30 dB	14.4 dB (≈ 1.1 dB below the 8×8 value)
Coded BER at 30 dB, hard / soft	5.2×10^{-2} / 1.1×10^{-2}
Soft-decision separation, 64-QAM region	5.05 dB at 10^{-1} (hard 27.09 dB, soft 22.04 dB); at 5×10^{-2} the hard curve does not reach the target

The columns of Table 5 mirror those of Table 4 so that the two summaries read side by side, and each entry again derives from the stored result file. The most consequential rows are the second and fourth: the least-squares floor and the permanently negative zero-forcing SINR are the two structural changes that the halved pilot density and the doubled inversion size introduced, and both feed directly into the comparison of Section 7.

7. Controlled Size Comparison: 8×8 against 16×16

Section 7 presents the controlled comparison of the two configurations: Section 7.1 the four overlay figures, and Section 7.2 the interpolated coded-BER crossings with the discussion of the central finding. Because the two runs differ in nothing but the antenna count, the overlays attribute every visible difference to array size alone. All four figures are generated by run_size_comparison.m from the two stored result files, and the numerical crossings of Table 6 are interpolated on the logarithmic BER axis from the same files by equation (13).

7.1 Overlay figures

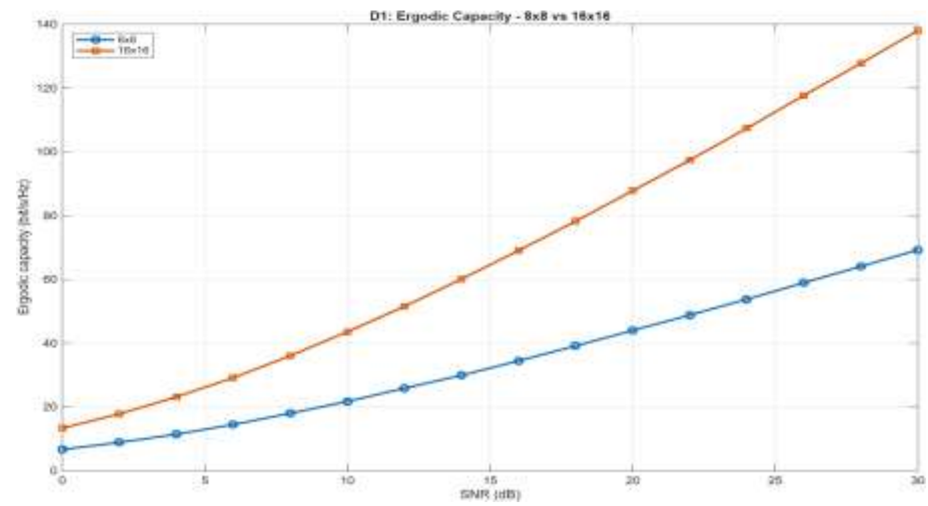

Figure 26. Ergodic capacity overlay of the two array sizes.

Figure 26 shows the 16×16 curve running at almost exactly twice the 8×8 curve across the entire sweep, 138.0 against 69.3 bit/s/Hz at 30 dB, confirming from executed Monte Carlo data that spatial-multiplexing capacity scales with the minimum of the transmit and receive antenna counts [5]. This is the return side of the trade that the remaining three figures price.

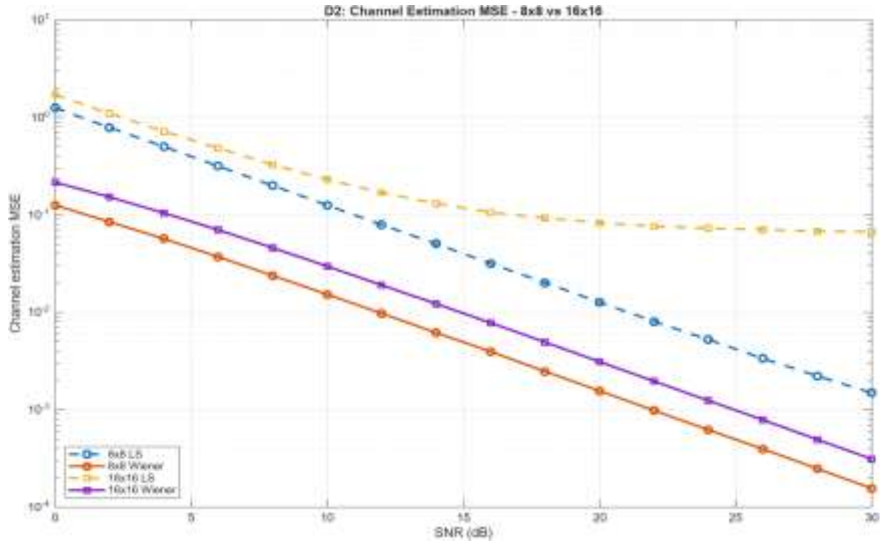

Figure 27. Channel-estimation mean-square error overlay for both estimators at both sizes.

Figure 27 presents four curves, correctly ordered throughout, telling the estimation story in one frame. Both 16×16 curves start above their 8×8 counterparts, the direct cost of halving the pilot density on the fixed subcarrier grid. The decisive feature is the divergence at high SNR: the 16×16 least-squares curve bends onto its interpolation floor while the 16×16 Wiener curve keeps the same clean slope as both 8×8 curves. The advantage of statistical estimation grows from roughly one decade at the smaller array to roughly two and a half decades at the larger one, which states quantitatively that channel statistics become more valuable, not less, as pilots become scarce.

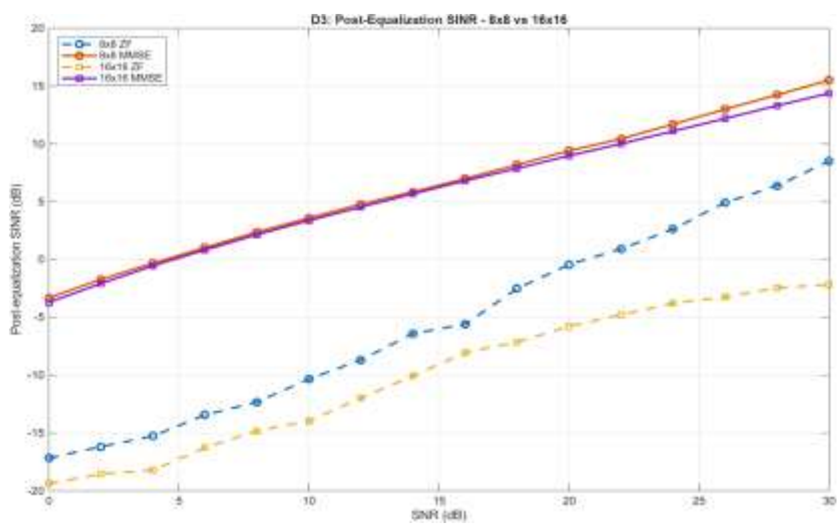

Figure 28. Post-equalization SINR overlay for both detectors at both sizes.

Figure 28 shows the two MMSE curves lying close to one another, separated by about 1.1 dB at the top of the sweep, while the two zero-forcing curves are split by up to about eleven decibels at high SNR and the 16×16 zero-forcing curve never reaches a positive output SINR.

Regularized detection is nearly indifferent to the array doubling; exact inversion pays for it heavily, both through its own conditioning and through the estimation floor it inherits from Figure 27.

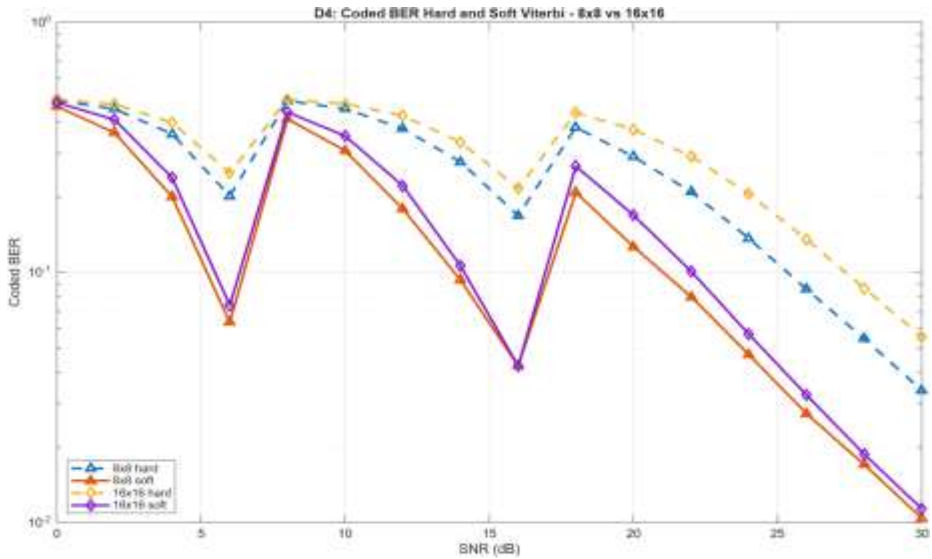

Figure 29. Coded BER overlay, hard- and soft-decision Viterbi at both sizes.

Figure 29 shows the two hard-decision curves clearly separated by array size, the 16×16 curve shifted right by two to three decibels, while the two soft-decision curves nearly coincide across the 64-QAM region and meet at the same one-percent endpoint. The sawtooth valleys align across all four curves at 6 and 16 dB, confirming the shared adaptive-modulation schedule. Read at a coded bit error rate of one in ten within the 64-QAM region, the separation between the soft-decision and hard-decision paths is 4.07 dB at 8×8 and 5.05 dB at 16×16, so the soft-decision advantage grows with array size.

7.2 Crossings and discussion

Table 6. Coded-BER crossings in the 64-QAM region by equation (13), interpolated on the logarithmic axis from the stored result files.

Configuration	Target BER	Hard-decision crossing	Soft-decision crossing	Soft gain
8×8	5×10^{-2}	28.06 dB	23.76 dB	4.30 dB
8×8	10^{-1}	25.10 dB	21.03 dB	4.07 dB
16×16	5×10^{-2}	not reached	24.45 dB	not reported
16×16	10^{-1}	27.09 dB	22.04 dB	5.05 dB

The first two columns of Table 6 identify the configuration and the target error rate, the middle columns state the interpolated SNR at which each decoder's coded curve crosses that target, and the final column states their separation, the measured soft-decision gain. One entry deserves explicit comment. At the stricter 5×10^{-2} target the 16×16 hard-decision curve ends the sweep at 5.2×10^{-2}, marginally above the target, so the analysis reports the crossing as not reached rather than extrapolating beyond the measured data. The refusal is itself a finding: hard-decision decoding at the doubled array requires more than 30 dB to reach an operating point that the smaller array reached at about 28 dB, while soft decoding reaches it at 24.45 dB regardless.

Taken together, the four overlays support a single thesis. Doubling a square array doubles the capacity on offer, but under linear detection the price is paid asymmetrically: interpolation-based estimation hits a floor, exact-inversion detection loses its remaining margin, and hard decisions surrender additional decibels, while their statistical counterparts, Wiener estimation, unbiased MMSE detection, and calibrated soft decoding, hold their performance nearly unchanged. A growth of the soft-decision separation with array size has a clean mechanism: a larger linear-detected array produces a wider spread of per-stream output SINRs, and log-likelihood ratios scaled by the per-stream effective noise variance of equation (11) are exactly the currency in which that spread is exploitable, whereas hard decisions discard it. The widening gap between offered capacity and linear-receiver performance is also the standard motivation for asymmetric many-antenna configurations [6], [7], in which excess receive dimensions restore the conditioning that square arrays lack.

8. Limitations and Future Work

The declared scope simplifications remain open for inspection. The Doppler model rotates all taps under one common phase together with per-pair random initial phases rather than independent per-path Doppler spectra, so the channel evolves coherently across paths; the timing estimate feeds the timing statistics and the carrier recovery but not the demodulation window, so plateau wander never propagates into bit errors; the reported SINR is a reference-based measure rather than a decomposed interference-versus-noise split; and throughput is a goodput proxy rather than a measured block-delivery rate. None of these affects the comparative conclusions of Section 7, which rest on differences between configurations evaluated under identical assumptions, but each bounds the absolute claims. The current implementation performs open-loop threshold-based adaptive modulation using the configured operating SNR: the convolutional-code rate remains fixed at 1/2, CQI is not generated, and modulation selection is not driven by receiver-estimated effective SINR or a block-error-rate target; these capabilities are developed separately as an advanced CQI-based adaptive modulation and coding extension. Further natural extensions follow the same discipline: a sum-of-sinusoids per-path Doppler model, sphere or successiveinterference-cancellation detection to price the linear-detection assumption itself, and the asymmetric many-antenna regime in which excess receive dimensions restore the conditioning that square arrays lack.

9. Conclusion

Two coordinated MATLAB studies were developed and verified within one PHY/modem simulation package: an OFDM synchronization study covering timing and two-stage carrier-frequency-offset recovery, and an 8×8/16×16 MIMO-OFDM link-level study covering channel estimation, spatial multiplexing, linear detection, threshold-based adaptive modulation, and fixed-rate hard- and soft-decision channel decoding. Both were implemented in MATLAB with full algorithm-to-code traceability, executed at two array sizes under otherwise identical conditions, and verified through an execution-first discipline that surfaced and eliminated four genuine defects. The synchronization subsystem recovers essentially ideal performance, with the recovered BER curve overlapping the genie reference and a carrier-offset RMSE of one thousandth of a subcarrier spacing at 30 dB. The controlled size study quantifies, from stored result files, how each receiver algorithm class responds to an array doubling: capacity doubles from 69.3 to 138.0 bit/s/Hz, the least-squares estimator falls onto an interpolation floor while the Wiener estimator retains its slope and widens its advantage to two hundredfold, zero forcing never reaches positive output SINR while unbiased MMSE concedes about 1.1 dB, and the separation between the soft-decision and hard-decision paths, measured from the common unbiased stream at a coded BER of one in ten, is 4.07 dB at 8×8 and grows to 5.05 dB at 16×16. The central finding is thereby stated quantitatively: the value of statistical processing, correlation-aware estimation, regularized detection, and reliability-calibrated soft decoding, grows with array size, and the parts of the receiver that discard statistical information absorb the entire penalty of the larger array.

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